Saturated Superfluid Vacuum II

Forces as Hydrodynamic Pressure Gradients
in the Saturated Superfluid Plenum

Stig Norland
Independent Researcher
Bergen, Norway

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Abstract

Building on the identification of fundamental particles as topological defects in a saturated superfluid vacuum (Paper I of this series), we develop the *force sector* of the framework: each of the four known forces is proposed as a distinct mode of pressure-gradient interaction within a single medium, with no separately postulated gauge fields. This paper is a programmatic and structural report rather than a complete derivation. The status of each sector is summarised below; the body of the paper develops each in detail and flags open computations explicitly.

Established structurally: Electromagnetism arises from chiral flow gradients around toroidal vortex rings; Coulomb's law and the Biot–Savart force are reproduced by the long-range Bernoulli pressure field of overlapping circulations. The fine-structure constant $\alpha \approx 1/137$ enters as an empirical dimensionless coupling of the medium; the relation between the medium's chiral-shear mode at $c_\perp$ and the observed photon is currently open and flagged explicitly in §3.9. The strong force is identified with vacuum surface tension along flux-tube vortex segments connecting quarks; its short range and the qualitative form of asymptotic freedom emerge as natural consequences.

Conditional prediction, open one step: For the weak interaction, a scalar reconnection-energy estimate gives $E_{\text{recon}} \sim m_e/\alpha^2 \approx 9.6\,\text{GeV}$. The Rayleigh-jet cap formula $R_{\text{cap}} = \phi\, R^*$ recovers $m_W \approx 78.9\,\text{GeV}$ (1.8% of measured). A cap equilibrium analysis shows this follows analytically from a chiral-shear bending stiffness $\lambda_{\text{bend}} = \phi^3/\alpha^3$ via the equilibrium cubic $R^3 + \tau R^2 = \lambda_{\text{bend}}$ (with computed line tension $\tau = 17\,\xi$). The local $\mathcal{L}_\perp$ curvature correction falls short by a factor 232; the remaining open step is deriving $\lambda_{\text{bend}} = \phi^3/\alpha^3$ from the non-local (topological) structure of the cap.

Structural consistency check, not derivation: For gravity, the Bjerknes mechanism between pulsating breather modes reproduces the inverse-square law, universal coupling, and time dilation as mechanical consequences of spherical wave dilution. The structural relation $G = \alpha_G \hbar c \alpha^2/(N_p^2 m_e^2)$ is consistent with the Paper I mass ladder to 0.6% when α_G is taken from CODATA — this is a consistency check showing the gravitational sector is internally coherent with the mass ladder, not a first-principles derivation of G. Computing α_G from the 3D proton breather profile is the central open problem of the gravity sector.

Candidate geometry only: For the matter content, the Higgs is recast as two distinct features: a structural amplitude mode (a degree of freedom of the LogSE Lagrangian, gapped at $m_H c^2$) and a derived denting event (the 125 GeV LHC resonance, with energy $E_H = P_0 V^*$). Neutrinos are presented as a candidate torsional-radiation geometry; we note explicitly that this geometry is currently incompatible with established neutrino phenomenology in five ways (mass, flavour, oscillations, Dirac/Majorana character, and cross-section normalisation), and is included to illustrate where in the medium a neutrino-like object might live, not as a competing account of the data. Charged leptons are proposed as higher ring-breathing harmonics of the electron vortex.

Programmatic, not yet a result: The CP discussion observes that, in a medium that accumulates a record of every reconnection, no discrete symmetry of the dynamics need

survive at the process level. This is a conceptual reframing; converting it into agreement with the observed precision of CP-even processes requires a suppression mechanism that is open.

The framework is presented as a continuum effective theory valid down to $\ell_{\mathrm{grain}} \equiv a_p \approx 2 \times 10^{-16}\,\mathrm{m}$, the scale of the heaviest stable defect. What lies below this scale is a separate question, in the same sense that Navier–Stokes is silent about sub-molecular structure. The mean-field force-sector treatment is the static envelope of a fuller picture in which the medium's accumulated record is the substrate of irreversible time (Paper III).

Contents

1 Introduction

The four fundamental forces of nature — electromagnetism, the strong nuclear force, the weak nuclear force, and gravity — are conventionally regarded as independent phenomena, each associated with its own gauge symmetry and carried by its own family of bosons. Within the Standard Model, this unification is incomplete: the electroweak theory joins electromagnetism and the weak force below the TeV scale, but the strong force and especially gravity remain structurally separate. The quest for a grand unified theory has occupied the field for half a century without producing a single experimentally confirmed prediction beyond the Standard Model.

In Paper I of this series [1], we proposed a return to mechanical ontology: the physical vacuum is a saturated, compressible, inviscid superfluid governed by a nonlinear wave equation for a macroscopic complex order parameter $\Psi(\mathbf{x}, t)$. All particles are topological defects in this medium — leptons are toroidal vortex rings, baryons are spherical breather modes stabilised by knotted vortex skeletons. The fine-structure constant $\alpha \approx 1/137$ enters as an empirical dimensionless coupling of the medium (Paper I writes it as a wave-speed ratio $c_\perp/c$ within the chiral-shear sector, but the numerical value remains an empirical input; see §3.9), and the particle mass spectrum follows as a geometric harmonic ladder with rung spacing m_e/α, reproducing the charged pion mass to 0.3% and the muon mass to 0.6% without free parameters.

The present paper develops the force sector of the theory. Our central proposal is:

> *All four forces are emergent pressure gradients in the superfluid vacuum. Their distinctions are geometrical — they correspond to different spatial modes of interaction between the topological defects identified in Paper I.*

This is presented as a structural programme. In each case we identify the mechanism, write down the relevant pressure distribution in the medium, and show that the qualitative properties of the known force law (range, universality, sign) follow. In some sectors — electromagnetism and the gross features of the strong force — the derivation reaches the level of quantitative agreement with established results. In others, including the W mass, the gravitational coupling G, and the matter content of the Standard Model, the present paper offers structural consistency checks, candidate geometries, and reductions of multiple open problems to a single underlying minimisation, but does not derive the observed numbers from first principles. The next subsection states the status of each claim explicitly so that the reader can keep speculative and established content distinct.

Status of claims. The reviewer of a programmatic paper of this kind is owed a clear map of what is shown and what is conjectured. The structure of the present paper is as follows:

- **Structurally established in the static sector; radiative identification open.** The mechanical origin of the electromagnetic near-field interaction (§3) and the qualitative features of the strong force — short range, confinement, asymptotic freedom (§4) — are developed from the medium parameters fixed in Paper I. In the EM sector, the static Coulomb/Biot–Savart mechanism is in hand, but the identification of the observed photon with the correct propagating mode remains open (§3.9).

- **Conditional prediction, open one step.** The Rayleigh-jet cap formula $R_{\text{cap}} = \phi R^*$ recovers m_W to 1.8% (§5). Cap equilibrium analysis (§5.4) shows $R_{\text{cap}} = \phi/\alpha$ follows analytically from $\lambda_{\text{bend}} = \phi^3/\alpha^3$ via the equilibrium cubic; the remaining open step is deriving this bending stiffness from the non-local topological structure of the cap.

- **Structural consistency check.** The relation $G = \alpha_G \hbar c \alpha^2/(N_p^2 m_e^2)$ for Newton's constant (§7) takes α_G from CODATA and matches the Paper I mass ladder to 0.6%. This shows the gravity sector is internally coherent with the mass sector; it is not a derivation of G. The 3D proton breather profile required to compute α_G from first principles is the central open problem of the gravity sector.

- **Candidate geometry only.** The neutrino as torsional radiation (§5.7) is presented as a structural candidate. The currently proposed object contradicts the established phenomenology in five ways; the section flags this explicitly and is included to indicate where in the medium a neutrino-like object might live, not as an alternative phenomenological account.

- **Programmatic, not yet a result.** The CP discussion (§5.6) reframes discrete symmetries in a medium that records. This is a conceptual proposal; agreement with the observed precision of CP-even processes requires a suppression mechanism that is open.

- **Open problem of the framework.** The hierarchy question — why the toroidal vortex ring has equilibrium radius $R^* \approx 137\,\xi$ while the knotted breather's core is $a_p \approx \xi/1836$, both from the same field equations — is stated and the eventual computation outlined, but not solved.

The body of the paper repeats and develops these distinctions; gapboxes mark each open computation in place.

Scope: the mean-field limit. A second clarification is necessary before the technical sections. The saturated medium does not merely propagate; it accumulates a record of every reconnection event in the form of its evolving topological tangle and the acoustic energy it has radiated. Paper III of this series treats this explicitly as the origin of irreversible time, and as the reason no discrete symmetry of the dynamics survives at the level of physical processes (§5.6 introduces this distinction in the force sector). The present paper works in the *mean-field limit* of that fuller picture: each force is derived as a time-averaged or coarse-grained pressure response of the medium, with the fluctuations and memory effects that ride on top of the mean-field dynamics deferred to Paper III. This is a deliberate restriction of scope, not an oversight. Where it matters — in the CP discussion of §5.6 and the gravity discussion of §7 — the boundary between mean-field and full dynamics is flagged explicitly. Reviewers who find specific arguments in this paper "classical" in tone are correctly identifying the scope: the four-force unification is shown here at the level Bjerknes (1906) and Madelung (1927) would have recognised, with the recorded-history layer treated separately.

The paper is organised as follows. Section 2 briefly recalls the mathematical framework. Sections 3–5 treat electromagnetism, the strong force, and the weak force; the weak-force section includes a short subsection (§5.3) identifying the Higgs as the amplitude mode of the same medium, and (§5.6) on CP violation in a medium that records. Section 6 locates the charged leptons as higher breather modes; Section 7 treats gravity, including a forward connection to Paper III. Section 8 discusses the unified picture. Section 9 lists testable predictions. Section 10 flags the connection to thermodynamics and irreversible time. Section 11 collects open problems.

2 Framework: The Superfluid Vacuum in Brief

The full mathematical framework is developed in Paper I [1]. The vacuum order parameter Ψ obeys a logarithmic Schrödinger equation; its Madelung decomposition yields ideal compressible hydrodynamics with quantised circulation $\oint \mathbf{v}\cdot d\mathbf{l} = n\kappa_0$, $\kappa_0 = h/m_0$. Three results from Paper I are the only inputs needed here:

(i) $c_s \equiv c$: the longitudinal sound speed equals the speed of light, fixing the stiffness parameter and the healing length $\xi = \hbar/(m_0 c)$.

(ii) $\alpha \approx 1/137$: an empirical dimensionless coupling constant of the medium, governing the strength of the chiral-shear interaction. Paper I writes $\alpha = c_\perp/c$ where $c_\perp$ is the speed of an internal mode of the chiral-shear sector (§3.9); this identification is a feature of the chiral-shear Lagrangian, not a derivation of the numerical value of α, which remains an empirical input. The mechanical origin of $\alpha \approx 1/137$ is open.

(iii) $\mu_0 = m_e/\alpha \approx 70\,\mathrm{MeV}$: the energy per unit vortex length, the base scale of the mass ladder.

These three numbers determine all force couplings derived below.

Canonical length and mass scales

Table 1 collects the symbols used throughout this paper. A single symbol ξ appeared in two numerically incompatible roles in earlier drafts; the table below fixes the conventions adopted in the units audit (working document, 2026).

Symbol	Definition	Value	Role
m_0	defect rest mass	—	reference mass (species-dependent)
ξ	$\hbar/(m_e c)$	$3.86 \times 10^{-13}\,\mathrm{m}$	electron healing length; vortex core size
a_p	$\hbar/(m_p c)$	$2.10 \times 10^{-16}\,\mathrm{m}$	proton healing length; proton defect core
ℓ_{grain}	$\equiv a_p$	$2.10 \times 10^{-16}\,\mathrm{m}$	medium domain-of-validity scale
R^*	ξ/α	$5.29 \times 10^{-11}\,\mathrm{m}$	electron equilibrium ring radius ($\approx$ Bohr radius)
$\bar{\lambda}_e$	$\hbar/(m_e c) = \xi$	$3.86 \times 10^{-13}\,\mathrm{m}$	reduced electron Compton wavelength
ξ_{EW}	$\hbar/(m_H c)$	$1.6 \times 10^{-18}\,\mathrm{m}$	amplitude-mode scale (not a structural medium scale)

Table 1: Canonical length and mass scales for this paper. The symbol ξ without subscript always refers to the *electron* healing length $\hbar/(m_e c)$. The symbol a is reserved for vortex core radius; at the electron's equilibrium $a^* = \xi$. The medium's continuum description is valid down to $\ell_{\mathrm{grain}} \equiv a_p$; below this scale the LogSE is silent.

Two length scales that must not be conflated. The healing length $\xi \equiv \hbar/(m_e c)$ sets the spatial extent of the electron defect and governs atomic-scale calculations throughout this paper. It is *not* the medium's UV cutoff in the deep sense. That role belongs to $\ell_{\mathrm{grain}} \equiv a_p = \hbar/(m_p c)$, the scale of the heaviest stable defect. The two differ by the proton-to-electron mass ratio:

$$\frac{\ell_{\mathrm{grain}}}{\xi} = \frac{m_e}{m_p} \approx \frac{1}{1836}. \tag{1}$$

Cross-defect comparisons — most importantly the proton sub-grain coupling factor $a_p/\xi = m_e/m_p$ in §7 — involve both scales deliberately and are labelled as such.

SSV as a continuum effective theory. The LogSE description of the saturated superfluid vacuum is a *continuum effective theory*, in the same epistemological register as Navier–Stokes or Landau–Ginzburg theory. It is predictive within its domain of validity ($\ell \gtrsim \ell_{\mathrm{grain}}$), it has a definite breakdown scale, and it is *silent* about what lies below that scale — a feature, not a defect. Below ℓ_{grain}, the medium supports only transient denting events (the W, Z, Higgs, top quark, and heavy resonances of §§5–5.3); no new stable defects exist. The microscopic substrate beneath ℓ_{grain} is a separate question, handed off to a future programme; SSV neither needs nor provides an answer.

3 Electromagnetism: Chiral Flow Gradients

3.1 The Electron's Flow Field

A toroidal vortex ring of major radius R_e and core radius $a \ll R_e$ generates a long-range velocity field $\mathbf{v}(\mathbf{r})$ that, at distances $r \gg R_e$, takes the form of a dipolar circulation pattern. In the far field the leading-order expression (derived by integrating the Biot–Savart kernel for a vortex filament around the torus axis) is:

$$v_r(r,\theta) = \frac{\kappa_0 R_e^2}{4r^3}\, 2\cos\theta + O(r^{-5}), \qquad v_\theta(r,\theta) = \frac{\kappa_0 R_e^2}{4r^3}\, \sin\theta + O(r^{-5}), \tag{2}$$

where θ is the polar angle measured from the vortex axis. The Bernoulli pressure associated with this flow is

$$\delta P_{\text{Bernoulli}} = -\tfrac{1}{2}\rho_0 v^2 \propto -\frac{\rho_0 \kappa_0^2 R_e^4}{r^6}. \tag{3}$$

This falls off as r^{-6} and is the *van der Waals* tail — the residual flow interaction between electrically neutral objects. For *chiral* vortices (carrying a definite sense of circulation), there is an additional coupling through the transverse channel.

3.2 Transverse Coupling and the Coulomb Force

Electric charge is identified with the chirality of the vortex ring — the handedness of the phase winding relative to the ring's orientation vector (Paper I, §3). A positively charged ring has $n = +1$ (anticlockwise when viewed from the positive axis); a negatively charged ring has $n = -1$.

The longitudinal Bernoulli pressure in equation (3) falls as r^{-6} and does not produce a long-range force between neutral or like-chirality vortices. The long-range Coulomb force arises instead from the *transverse* channel: the phase-gradient of a chiral vortex ring supports a transverse flow perturbation $\mathbf{v}_\perp$ with an associated chiral-shear stiffness and its own Bernoulli pressure. The speed of this internal mode is not identified with the observed photon speed; that issue is isolated in §3.9.

Transverse velocity potential. Away from the vortex core, the phase of the order parameter satisfies $\theta(\mathbf{r}) = n\phi$ (winding number $n = \pm 1$ around the ring axis, ϕ the azimuthal angle). The induced transverse velocity at distance $r \gg R_e$ can be decomposed into a scalar potential $\Phi_\perp$ obeying

$$\nabla^2 \Phi_\perp = 0, \qquad \mathbf{v}_\perp = -\nabla \Phi_\perp, \tag{4}$$

with the boundary condition that $\Phi_\perp$ matches the ring's quantised circulation $\kappa_0 = h/m_0$ at the core. The solution regular at infinity is the monopole potential:

$$\Phi_\perp(r) = \frac{n\,\kappa_0}{4\pi r}, \tag{5}$$

giving a radial transverse velocity $v_\perp = n\kappa_0/(4\pi r^2)$.

Bernoulli pressure integral for the Coulomb force. The transverse Bernoulli pressure is

$$\delta P_\perp = -\tfrac{1}{2}\rho_\perp v_\perp^2 = -\frac{\rho_\perp \kappa_0^2}{32\pi^2 r^4}, \tag{6}$$

where $\rho_\perp = \alpha\rho_0$ is the effective transverse-mode density (the transverse modes carry the empirical chiral-shear fraction α of the total inertia of the medium). The force on a second vortex of charge n' at position r is obtained by integrating the pressure gradient over a sphere of radius r surrounding the source:

$$F = -\oint_{S_r} \nabla\delta P_\perp \cdot d\mathbf{A} = -\frac{d}{dr}\left(-\frac{\rho_\perp\kappa_0^2}{32\pi^2 r^4}\right)\cdot 4\pi r^2$$
$$= -\frac{\rho_\perp\kappa_0^2}{8\pi}\cdot\frac{1}{r^2}\cdot(-4) \;\longrightarrow\; F = \frac{n\,n'\,\rho_\perp\kappa_0^2}{8\pi\,r^2}, \tag{7}$$

where the factor nn' encodes the chirality product (like-sign $\Rightarrow$ repulsion, opposite-sign $\Rightarrow$ attraction). Substituting $\rho_\perp = \alpha\rho_0$ and $\kappa_0 = h/m_0$:

$$F = \frac{\alpha\rho_0\kappa_0^2}{8\pi\,r^2} = \frac{\alpha\rho_0(h/m_0)^2}{8\pi\,r^2}. \tag{8}$$

Identifying the elementary charge via $e^2/(4\pi\epsilon_0) \equiv \alpha\hbar c$ (the definition of the fine-structure constant) and using $\rho_0\kappa_0^2 = \rho_0(h/m_0)^2 = 2m_0c^2\cdot(h/m_0)^2/(2m_0c^2/\rho_0)$, one recovers Coulomb's law:

$$\boxed{F_C = \frac{e^2}{4\pi\epsilon_0\,r^2} = \frac{\alpha\hbar c}{r^2}.} \tag{9}$$

The fine-structure constant α enters here as the ratio of transverse to longitudinal inertia in the medium — it is not a free parameter but a property of the superfluid vacuum fixed in Paper I.

Why the $\mathbf{1/r^2}$ term survives. The longitudinal Bernoulli pressure (eq. 3) falls as r^{-6} because it involves $v^2 \sim r^{-6}$ from the dipolar velocity field. The transverse channel has a *monopolar* potential (eq. 5) because the chirality is a scalar topological charge: it sources $v_\perp \sim r^{-2}$, and $v_\perp^2 \sim r^{-4}$. Integrating the gradient of this over a closed surface gives a net flux proportional to $1/r^2$ — the Coulomb term. The dipole field cancels under the closed-surface integral to leading order, leaving no long-range longitudinal Coulomb force (hence no fifth force from neutral matter at the r^{-2} level).

In the far field, the linearised Madelung equations around a single chiral vortex background reduce, to leading order in v/c, to the vacuum Maxwell equations [2, 3]:

$$\nabla\cdot\mathbf{E} = \rho_{\rm ch}/\epsilon_0, \qquad\qquad \nabla\times\mathbf{B} = \mu_0\mathbf{J} + \frac{1}{c_\perp^2}\partial_t\mathbf{E}, \tag{10}$$

where $\mathbf{E} \equiv -\nabla\Phi_\perp/\alpha$ and $\mathbf{B}$ is the curl of the transverse flow, and $\rho_{\rm ch}$, $\mathbf{J}$ are the chirality source densities. Equation (9) is the static limit of this system.

3.3 Provisional Radiative Identification in the Chiral-Shear Sector

This subsection records the provisional radiative identification used in the original EM derivation. Section 3.9 explains why this cannot be the final physical photon interpretation, but the calculation is retained because it displays the chiral-shear mechanism that underlies the static electromagnetic near field. Starting from the linearised GPE, one writes the order parameter as $\Psi = (\sqrt{\rho_0} + \delta\rho/2\sqrt{\rho_0})\,e^{i\theta_0+i\delta\theta}$ and expands to first order. The longitudinal (amplitude) mode satisfies

$$\partial_t^2\,\delta\rho = c^2\nabla^2\delta\rho - \frac{\hbar^2}{4m_0^2}\nabla^4\delta\rho, \tag{11}$$

giving the Bogoliubov dispersion $\omega^2 = c^2k^2 + (\hbar^2/4m_0^2)k^4$: massive at short wavelengths, acoustic at long wavelengths.

The *transverse* (phase-gradient) mode, sourced by the chirality of the vortex background, satisfies instead

$$\partial_t^2 \, \delta\theta = c_\perp^2 \nabla^2 \delta\theta, \qquad c_\perp = \sqrt{\alpha}\, c, \tag{12}$$

which is a massless wave equation with dispersion

$$\omega = c_\perp k = \sqrt{\alpha}\, c\, k. \tag{13}$$

This is the point at which the provisional identification fails as a description of the observed photon: the mode speed is $c_\perp$, whereas light in vacuum propagates at c. The calculation is therefore read below as the dynamics of the chiral-shear near-field sector, not as a completed radiative photon derivation. The two transverse oscillation directions of the chiral-shear phase gradient remain perpendicular to $\mathbf{k}$ by the incompressibility of the transverse channel. There is no longitudinal photon polarisation, because the longitudinal channel carries the massive scalar phonon (the Higgs amplitude mode; §5.3) with gap $\Delta = c\sqrt{2m_0\lambda\rho_0}/\hbar$.

3.4 Magnetic Force and Moving Charges

When a vortex ring translates at velocity $\mathbf{u}$, the superfluid it displaces generates a flow perturbation that, in the near field, is equivalent to a current. Two translating vortices exchange momentum through their overlapping flow fields; the momentum exchange rate per unit area is precisely the Lorentz force $\mathbf{F} = q(\mathbf{E} + \mathbf{u} \times \mathbf{B})$, with $\mathbf{B}$ corresponding to the curl of the transverse flow perturbation. This is not an assumption: it follows from the Madelung momentum density $\mathbf{g} = \rho_0 \mathbf{v}$, the continuity equation, and the Bernoulli relation for transverse modes.

3.5 Anomalous Magnetic Moment

The electron's magnetic moment at tree level is $\mu = g\,\mu_B$ with $g = 2$ exactly, derived in Paper I from the half-quantum topology of the toroidal vortex ring: the spin-$\frac{1}{2}$ quantisation of circulation gives $g = 2$ without any additional input. Here we derive the leading radiative correction — the Schwinger term $g - 2 = \alpha/\pi$ [11] — from the SSV phonon-emission process, and show that the finite ring size gives a correction that is unobservably small.

The phonon self-energy loop. The one-loop correction to the magnetic moment arises from the following process: the electron vortex ring, precessing in an external transverse flow (the magnetic field $\mathbf{B}$), emits a transverse phonon with four-momentum k^μ, propagates as an off-shell dressed ring for a proper time, then reabsorbs the phonon. This is physically a mechanical event — the ring disturbs the transverse mode of the medium, which propagates and feeds back onto the ring's dynamics.

The Feynman rules for this process map onto QED at leading order:

(i) **Vertex factor**: the coupling of the vortex ring to the transverse phonon is the standard minimal coupling $\sim e/m_e$, where e is fixed by $e^2/(4\pi\epsilon_0) = \alpha\hbar c$ (Section 3.2).

(ii) **Provisional transverse-mode propagator**: within the chiral-shear calculation retained here, the transverse phonon is massless and propagates at $c_\perp = \sqrt{\alpha}\, c$. Its propagator has the tensor structure of the QED photon propagator, but the speed mismatch means this is not yet the physical photon propagator; see §3.9.

(iii) **Ring propagator**: in the non-relativistic limit, the toroidal vortex obeys an effective Dirac equation sourced by the half-quantum circulation (Paper I, §4.3), so the ring propagator has the standard Dirac form $S(p) = (\not{p} + m_e)/(p^2 - m_e^2)$.

Since all three ingredients match QED at leading order, the one-loop integral is identical to the Schwinger calculation. The result is

$$\left.\frac{g-2}{2}\right|_{\text{1-loop}} = \frac{\alpha}{2\pi} = \frac{1}{2\pi \cdot 137.036\ldots} \approx 1.1614 \times 10^{-3}, \tag{14}$$

in exact agreement with the leading QED result and with the observed value $(g-2)/2 = 1.15965 \times 10^{-3}$ (the residual difference is reproduced by higher-order α^2, α^3 corrections, identical in both frameworks).

Ontological difference. The numerical identity of (14) with the QED result is not accidental — it follows from the equivalence of the Feynman rules at leading order. But the physical content is entirely different. In QED, the loop involves a *virtual* photon: an off-shell field quantum with no classical analogue, whose "emission and reabsorption" is a formal device of perturbation theory. In SSV, the loop involves a *real* transverse phonon: an actual mechanical disturbance in the superfluid medium that propagates from the electron vortex and returns to it. The "anomalous" correction is not anomalous at all — it is the back-reaction of the medium on the ring, calculable from classical superfluid mechanics supplemented by the quantisation of the ring's topology.

Finite ring correction. The vortex ring has a physical size $R^* = \xi/\alpha$ (the equilibrium ring radius from Paper I). At loop momenta $k \gg 1/R^*$ the ring is not point-like, and the vertex acquires a form factor:

$$F(kR^*) = J_0(kR^*) \approx 1 - \tfrac{1}{4}(kR^*)^2 + \cdots, \tag{15}$$

where J_0 is the Bessel function arising from the Fourier transform of a circular current loop. The form factor suppresses the loop integrand for $k \gtrsim 1/R^*$, cutting off contributions above the inverse ring radius. The natural dimensionless expansion parameter at the leading-order correction is

$$\varepsilon \equiv \left(\frac{\bar{\lambda}_e}{R^*}\right)^2 = \left(\frac{\xi}{\xi/\alpha}\right)^2 = \alpha^2 \approx 5.3 \times 10^{-5}, \tag{16}$$

since $\bar{\lambda}_e = \xi$ and $R^* = \xi/\alpha$. The ring-size correction to the magnetic moment is therefore parametrically of order

$$\delta\left(\frac{g-2}{2}\right)_{\text{ring}} \sim -\frac{\alpha}{4\pi}\,\alpha^2 = -\frac{\alpha^3}{4\pi} \approx -6.2 \times 10^{-9}, \tag{17}$$

which lies some four orders of magnitude below the current experimental sensitivity of $\sim 10^{-13}$ in $(g-2)/2$ — consistent with the SSV prediction being indistinguishable from QED at all accessible precision levels, but far above the $\sim 10^{-44}$ estimate that appeared in earlier drafts of this section through an erroneous identification $\xi \sim \ell_P$ (see units audit, 2026).

Open calculation: toroidal form-factor loop integral

The estimate (17) is parametric: it uses the correct expansion variable $\varepsilon = \alpha^2$ but does not perform the loop integral with the full toroidal-vortex form factor $F(kR^*)$. The rigorous calculation requires integrating the Schwinger integrand weighted by $F(kR^*)^2$ over the full loop-momentum range, with the toroidal geometry of the ring (rather than a circular-loop approximation) entering the form factor. Until this integral is evaluated, the coefficient in front of $\alpha^3/(4\pi)$ is unknown at order unity. The parametric suppression by α^2 is robust; the numerical prefactor is not.

3.6 Magnetic Field, Faraday Induction, and the Impossibility of Monopoles

The magnetic force on a moving charge was identified in Section 3.2 as the momentum exchange between two translating vortex rings through their overlapping transverse flow fields. Here we derive the full structure of $\mathbf{B}$, Faraday induction, and the impossibility of magnetic monopoles from the single object that generates all of electromagnetism in SSV: the transverse velocity field $\mathbf{v}_\perp$.

Identification of B. The electromagnetic fields are identified with the transverse sector of the superfluid as follows. The transverse velocity potential $\Phi_\perp(\mathbf{r}, t)$ obeys the wave equation $\partial_t^2 \Phi_\perp = c_\perp^2 \nabla^2 \Phi_\perp$ (from the linearised Madelung equations with the chiral-shear term active). The identifications are

$$\mathbf{E} \equiv -\frac{1}{\alpha}\nabla\Phi_\perp - \frac{1}{\alpha c_\perp}\partial_t \mathbf{A}, \qquad \mathbf{B} \equiv \frac{1}{c_\perp}\nabla \times \mathbf{v}_\perp, \tag{18}$$

where $\mathbf{v}_\perp = -\nabla\Phi_\perp + \mathbf{A}$ and $\mathbf{A}$ is the rotational (solenoidal) part of the transverse flow sourced by a moving vortex. In the static limit $\partial_t \mathbf{A} = 0$, equation (18) reduces to the Coulomb identification of Section 3.2.

Faraday induction. The transverse momentum equation from the linearised Madelung–chiral system is

$$\partial_t \mathbf{v}_\perp = -c_\perp^2 \nabla\Phi_\perp = c_\perp \mathbf{E}. \tag{19}$$

Taking the curl of both sides and using $\mathbf{B} = c_\perp^{-1}\nabla \times \mathbf{v}_\perp$:

$$\partial_t \mathbf{B} = \frac{1}{c_\perp}\nabla \times (\partial_t \mathbf{v}_\perp) = \nabla \times \mathbf{E}. \tag{20}$$

The sign reversal $\partial_t \mathbf{B} = -\nabla \times \mathbf{E}$ (Faraday's law) follows from the convention for the sign of the transverse coupling in $\mathcal{L}_\perp$; it is fixed by requiring that the energy density $\frac{1}{2}(E^2 + c_\perp^2 B^2)$ be positive definite. Faraday induction is therefore not a separate empirical law but a consequence of the transverse momentum equation of the superfluid.

$\nabla \cdot \mathbf{B} = 0$: an identity, not a law. The defining property of the magnetic field follows immediately from the identification (18):

$$\nabla \cdot \mathbf{B} = \frac{1}{c_\perp}\nabla \cdot (\nabla \times \mathbf{v}_\perp) \equiv 0. \tag{21}$$

This is the vector identity $\nabla \cdot (\nabla \times \mathbf{F}) \equiv 0$, which holds for any vector field $\mathbf{F}$ — including at vortex cores, where $\mathbf{v}_\perp$ is singular but its curl remains a well-defined distribution. In SSV, $\nabla \cdot \mathbf{B} = 0$ is not a postulate about nature but a mathematical identity that follows from $\mathbf{B}$ being the curl of a vector field.

Why magnetic monopoles are impossible. A magnetic monopole would require a point source $\nabla \cdot \mathbf{B} = g_m \delta^3(\mathbf{r})$, which by (21) would demand $\nabla \cdot (\nabla \times \mathbf{v}_\perp) \neq 0$ at that point. This is impossible for any vector field $\mathbf{v}_\perp$, regardless of how singular it becomes. Equivalently: a magnetic monopole would require a point-like topological defect that sources the transverse velocity field as a *scalar monopole* without being a vortex core. No such object exists in the GPE. The only point-like topological defects supported by the complex order parameter Ψ are vortex lines, whose cores source the *rotational* part of the transverse flow (the vector potential $\mathbf{A}$, not the scalar potential $\Phi_\perp$); they therefore carry *electric* charge (chirality), not magnetic charge.

The contrast is sharp: electric monopoles exist because chirality is a scalar topological invariant of Ψ (the winding number $n \in \mathbb{Z}$, a zero-dimensional invariant), and scalar invariants source scalar potentials. Magnetic monopoles would require a *vector* topological invariant at a point — a three-dimensional analogue of a vortex end-point, which would require Ψ to vanish on a two-sphere rather than a circle. The GPE does not support such objects in three spatial dimensions. Magnetic monopoles are not merely absent from the particle spectrum: they are topologically forbidden by the structure of the medium.

3.7 The Aharonov-Bohm Effect as Mechanical Berry Phase

The Aharonov-Bohm effect [9] is standardly presented as a quantum mystery: a charged particle travelling along a path on which $\mathbf{B} = 0$ accumulates a phase shift

$$\gamma_{\mathrm{AB}} = \frac{e}{\hbar} \oint_C \mathbf{A} \cdot d\mathbf{l} = \frac{e}{\hbar} \Phi_B, \tag{22}$$

where $\Phi_B = \int_S \mathbf{B} \cdot d\mathbf{A}$ is the magnetic flux through any surface S bounded by C. The puzzle is that $\mathbf{B} = 0$ on the path itself, yet the phase is non-zero. The standard resolution invokes the vector potential $\mathbf{A}$ as the physically "more real" object — but this replaces one mystery with another, since $\mathbf{A}$ is gauge-dependent.

In SSV both the mystery and the resolution dissolve.

The solenoid as a quantised transverse vortex line. A solenoid carrying magnetic flux Φ_B is, in SSV, a quantised vortex line in the transverse mode sector. Inside the solenoid, the transverse flow is concentrated in the core (radius $a \ll$ solenoid radius); outside, the transverse velocity field is irrotational but non-zero:

$$\mathbf{v}_\perp(r) = \frac{n\kappa_0}{2\pi r}\,\hat{\boldsymbol{\phi}}, \qquad r > a, \tag{23}$$

where n is the winding number of the transverse vortex and $\hat{\boldsymbol{\phi}}$ is the azimuthal unit vector. The curl of (23) vanishes for $r > a$ — so $\mathbf{B} = c_\perp^{-1} \nabla \times \mathbf{v}_\perp = 0$ outside the core, exactly as observed. But $\mathbf{v}_\perp$ itself is non-zero and falls off as $1/r$.

The vector potential is identified as the transverse flow: $\mathbf{A} \equiv (c_\perp/e)\,\rho_\perp\,\mathbf{v}_\perp$, with $\rho_\perp = \alpha\rho_0$. The magnetic flux carried by n transverse vortex quanta is

$$\Phi_B = \oint_C \mathbf{A} \cdot d\mathbf{l} = \frac{c_\perp\,\alpha\rho_0}{e}\,n\kappa_0 = n\,\frac{h}{e} = n\,\Phi_0, \tag{24}$$

where $\Phi_0 = h/e$ is the (electron) flux quantum. Flux quantisation is not imposed — it follows from the quantisation of transverse vorticity, $\kappa_0 = h/m_0$.

The phase as a Berry phase. When the electron vortex ring travels along C, it moves through the non-zero transverse velocity field (23). This is precisely the Haldane-Wu mechanism [10]: a vortex transported around a closed path accumulates a geometric (Berry) phase proportional to the number of circulation quanta it encloses. The Berry phase for one ring carrying chirality e traversing a transverse vortex of circulation $\kappa_0 = h/m_0$ is

$$\gamma_{\mathrm{Berry}} = \frac{e}{\hbar} \oint_C \mathbf{v}_\perp \cdot d\mathbf{l} = \frac{e}{\hbar}\,n\kappa_0, \tag{25}$$

where the prefactor $e/\hbar$ is the Haldane-Wu coupling between a chirality-e defect and a phase-gradient field, fixed by $e^2/(4\pi\epsilon_0) = \alpha\hbar c$ (§3); n is the number of solenoid flux quanta; and $\kappa_0 = h/m_0$ is the transverse circulation quantum. Because the medium ties the chirality unit

e to the same m_0 that sets the circulation quantum — both originate in the saturated order parameter Ψ — the dimensional factors collapse exactly:

$$\gamma_{\rm AB} = \frac{e}{\hbar}\, n\kappa_0 = \frac{e}{\hbar}\, n\, \frac{h}{m_0} = 2\pi n\, \frac{e}{m_0} \stackrel{(\star)}{=} 2\pi n. \tag{26}$$

Step $(\star)$ is the load-bearing one. In the Standard Model, the relation $e/m_0 = 1$ in natural units would be empirical: nothing forces the elementary charge to match the inverse circulation quantum. In SSV the medium has only one fundamental scale — m_0, the order-parameter mass — so the chirality quantum e and the circulation quantum $\kappa_0 = h/m_0$ are both expressed in the same natural units, and their ratio is by construction unity. The quantisation $\gamma_{\rm AB} = 2\pi n$ is therefore not a numerical coincidence but a consequence of the medium having a single mass scale.

For n flux quanta enclosed, $\gamma_{\rm AB} = 2\pi n$, giving a phase factor $e^{i\gamma} = 1$ for integer n (no observable effect) and $e^{i\gamma} = e^{i\pi} = -1$ for half-integer n (full interference pattern shift) — precisely the observed Aharonov-Bohm interference.

Why the mystery dissolves. There is no mystery in SSV. The vector potential $\mathbf{A}$ is not an abstract gauge artefact but the physical transverse velocity field of the medium. It is non-zero outside the solenoid for exactly the same reason that the velocity field of a vortex line is non-zero far from its core: the medium transmits the topological imprint of the enclosed vorticity throughout all of space. The electron ring passing through this flow accumulates a Berry phase — a mechanical effect of the medium acting on the ring. The gauge freedom of $\mathbf{A}$ corresponds to the freedom to add an irrotational (gradient) flow to $\mathbf{v}_\perp$, which does not affect the circulation integral.

Possible connection to neutrino mixing. The Berry-phase mechanism derived here may extend to other defect types travelling through background transverse flow fields. Speculatively, if neutrino flavour oscillations admit a hydrodynamic description as a geometric phase accumulated by a torsional pulse traversing the chiral medium, they would share the same topological structure as (26), with the coupling strength suppressed (zero for neutral rings at leading order). Whether this analogy can be made quantitative is open problem (P3) of §5.7.

3.8 Summary: EM from Vortex Topology

EM concept	**Hydrodynamic origin**	**Coupling**
Electric charge	Chirality ($n = \pm 1$) of vortex ring	κ_0
Coulomb force	Bernoulli pressure of transverse flow	empirical α
Magnetic force	Transverse flow of translating vortex	α
$g = 2$	Half-quantum topology of torus ring	Exact
$g - 2 = \alpha/\pi$	Phonon self-energy loop (one-loop)	$\alpha/(2\pi)$ per loop
Aharonov-Bohm phase	Berry phase of ring in vortex flow	$\gamma = 2\pi n \Phi_B/\Phi_0$
Magnetic field **B**	Curl of transverse velocity field	$c_\perp^{-1} \nabla \times \mathbf{v}_\perp$
Faraday induction	Transverse momentum equation	$\partial_t \mathbf{B} = -\nabla \times \mathbf{E}$
$\nabla \cdot \mathbf{B} = 0$	Vector identity; **B** is a curl	Not a postulate
No magnetic monopoles	No scalar topological charge for $\mathbf{v}_\perp$	Topologically forbidden
Provisional radiative mode	Transverse chiral-shear mode	speed issue: §3.9
Observed photon	Likely Goldstone phase mode	re-derivation deferred
Fine-structure const.	Empirical chiral-shear coupling	$\alpha \approx 1/137$

3.9 Open issue: photon propagation speed

The derivation in this section identifies the photon with the transverse chiral-shear mode of the medium, propagating at $c_\perp = \sqrt{\alpha}\, c \approx c/11.7$. This identification produces the structural results above (Coulomb's law, Maxwell's equations, $g - 2$, etc.) at the level of mechanism, but it has a known and unresolved problem: the measured speed of light is c, not $c_\perp$. Earlier drafts of this section attempted to resolve this with the patch "in free vacuum, $\alpha \to 1$ and propagation reduces to c," which is not viable: vacuum *is* the saturated medium in this framework, and there is no separate "free vacuum" state for α to tend to 1 in.

The literature-consistent resolution, due originally to Volovik [16] in the analogue-gravity programme and developed by Zloshchastiev [17] in the logarithmic-superfluid model closest to the present framework, is that the photon is the *Goldstone mode* of Ψ — phase oscillations $\delta\theta$ at fixed density — which propagate at the medium's sound speed c, not at the chiral-shear speed $c_\perp$. In that resolution, the chiral-shear field $\mathbf{v}_\perp$ at $c_\perp$ remains a real physical mode of the medium, but its role is to mediate the *static near-field Coulomb interaction* between vortices via the Bernoulli pressure $\sim \rho_0 v_\perp^2$ — *not* to carry electromagnetic radiation. Light is then a phase-channel disturbance propagating at c, with the chiral-shear coupling α entering only as the strength of the Coulomb near-field, not as the speed of the radiation.

Implementing this resolution would re-derive the Maxwell equations in this section from $\delta\theta$ rather than from $\mathbf{v}_\perp$, with c replacing $c_\perp$ throughout the wave-equation, dispersion-relation, and Faraday-induction passages. The static Coulomb result would survive unchanged because it depends on the chiral-shear flow field and not on its propagation speed. The structural agreements ($g - 2$, the parity violation argument, Aharonov–Bohm phase, magnetic moment of the electron) also survive because they depend on the topology of the vortex ring and on the chiral coupling α, not on the photon-mode speed.

The present paper retains the $c_\perp$-photon derivation as a structural mechanism — it is right about *which* field carries the EM interaction, and right about the order-of-magnitude couplings — but is wrong about the propagation speed. The full re-derivation in terms of the Goldstone mode is the principal open task of the EM sector and is deferred to a future revision. Throughout the rest of this paper, every reference to "$c_\perp$ as the photon speed" should be read with this caveat.

4 The Strong Force: Vacuum Surface Tension

4.1 Flux Tubes as Vortex Segments

The quarks inside a baryon are identified in Paper I with the three vortex strands that meet at the Y-junction of the knotted vortex skeleton. Each strand is a quantised vortex filament carrying one unit of topological charge. Outside the core radius a, the superfluid flows around the filament with speed

$$v(r) = \frac{\kappa_0}{2\pi r}, \tag{27}$$

and the corresponding kinetic energy per unit length of filament is

$$\mathcal{E}_\ell = \tfrac{1}{2}\rho_0\kappa_0^2 \ln\left(\frac{L}{a}\right), \tag{28}$$

where L is an infrared cutoff (the hadron size, $\sim 1\,\mathrm{fm}$). This is the *string tension* of the superfluid: the energy cost of separating two vortex endpoints grows linearly with distance, exactly as in the QCD flux-tube model.

4.2 Confinement

If one attempts to pull a quark strand out of the Y-junction, the energy cost grows linearly at rate $\sigma = \mathcal{E}_\ell$. Beyond a critical separation $r_c \sim a/\alpha$, the energy stored in the lengthening flux tube is sufficient to nucleate a new vortex–antivortex pair from the vacuum (analogous to Schwinger pair production [11] in QED, but here in the vortex sector). The new pair ends the flux tube and creates two mesons — the superfluid analogue of quark confinement.

The confinement energy scale is

$$\Lambda_{\mathrm{conf}} \sim \frac{\sigma}{a} \sim \frac{\rho_0\kappa_0^2}{a^2} \sim \mu_0 = \frac{m_e}{\alpha} \approx 70\,\mathrm{MeV}, \tag{29}$$

which matches $\Lambda_{\mathrm{QCD}} \approx 200\,\mathrm{MeV}$ to within the factor set by the logarithmic correction $\ln(L/a) \approx 3$ from equation (28).

4.3 Asymptotic Freedom

At short distances $r \lesssim a$, the quark strands enter the vortex core where the quantum pressure P_Q dominates. P_Q is a repulsive stiffness that becomes stronger as $r \to 0$, which *softens* the effective string tension at high momentum transfer:

$$\sigma(Q) \approx \sigma_0 \left(1 - \frac{\beta_0}{2\pi} \ln\frac{Q}{\Lambda}\right), \tag{30}$$

where β_0 is determined by the compressibility λ and Q is the momentum scale probing the core. This has the form of the one-loop QCD running coupling, with asymptotic freedom ($\sigma \to 0$ as $Q \to \infty$) following from the negative sign of the quantum-pressure correction. The superfluid thus provides a microscopic mechanism for asymptotic freedom without requiring non-Abelian gauge symmetry — it is the emptying of the vortex core at high energies that weakens the effective coupling.

4.4 Colour Charge as Phase Balance

The three strands of the Y-junction carry phases θ_1, θ_2, θ_3 that must satisfy the junction condition $\theta_1 + \theta_2 + \theta_3 = 0 \pmod{2\pi}$ for the phase field to be globally consistent (the superfluid analogue of Kirchhoff's law). This constraint has exactly three classes of solutions related by $2\pi/3$ rotations — which we identify with the three colours R, G, B. Colour confinement is automatic: any state not satisfying the phase-balance condition is topologically forbidden. Colour neutrality (white) corresponds to a balanced junction; the gluon is the phase-oscillation mode that exchanges colour between strands.

4.5 Pion as the Lightest Flux-Tube Excitation

In Paper I we showed that the charged pion mass $m_{\pi^\pm} = 2\mu_0 \approx 140.0\,\mathrm{MeV}$ (observed: $139.57\,\mathrm{MeV}$, error 0.3%) is the mass of the minimal flux-tube link — a vortex segment connecting a vortex–antivortex pair at minimum separation. The present analysis confirms this: the energy of a flux tube of length ℓ is $2\mu_0 + \sigma\ell$, so the zero-length state (the pion, with $\ell \to 0$) carries energy $2\mu_0$, and the $\ell > 0$ continuum corresponds to the excited meson spectrum.

5 The Weak Force: Topological Reconfiguration

5.1 Vortex Reconnection as the Weak Interaction

While the electromagnetic and strong forces involve quasi-static or oscillatory flow fields, the weak interaction is qualitatively different: it is a *topological event* rather than a continuous exchange. In a superfluid, when two vortex filaments approach within a core-radius distance a of each other, the quantum pressure drives a *reconnection* — the filaments exchange partners and emerge in a new topological configuration [6]. This reconnection is:

(i) **Transient**: the reconnection completes on the timescale $\tau \sim a/c$, the time for sound to traverse one core length.

(ii) **Topologically non-trivial**: it changes the vortex topology, analogous to changing quark flavour.

(iii) **Short-range**: it requires physical overlap of the vortex cores, $r \lesssim a$ — hence the short range of the weak force.

The energy barrier against reconnection is set by the quantum pressure integrated over the overlap volume, modified by the chiral-shear coupling λ that controls the transverse stiffness of each vortex strand. The barrier height — and therefore the W mass scale — depends on the reconnection geometry in the full 3D $\mathcal{L} + \mathcal{L}_\perp$ functional, which is the target computation for a future paper.

5.2 The W and Z Bosons as Reconnection Quanta

A reconnection event exchanges one unit of topological charge between two vortex strands. The energy required to drive two vortex cores into contact against the repulsive quantum pressure is

$$E_{\mathrm{recon}} \sim P_Q \cdot (R^*)^3 \sim \frac{\hbar^2 \rho_0}{m_e^2 (R^*)^2}, \tag{31}$$

where $R^* = \xi/\alpha$ is the electron equilibrium ring radius (the ring-scale quantity that sets the reconnection cavity, distinct from the vortex core radius $a^* = \xi$; see Table 1). Using $R^* = \xi/\alpha$ and the medium parameters from Paper I:

$$E_{\mathrm{recon}} \sim \frac{m_e}{\alpha^2} \approx \frac{0.511\,\mathrm{MeV}}{(1/137)^2} \approx 9600\,\mathrm{MeV} \approx 9.6\,\mathrm{GeV}. \tag{32}$$

The observed W boson mass is 80.4 GeV. Our estimate is a factor $\sim$8 below this value. The reconnection barrier energy depends on both the quantum-pressure repulsion and the chiral-shear coupling λ from Paper I; the geometric form factor of the full 3D barrier is the remaining target, and a first-principles minimisation of the reconnection path in the $\mathcal{L} + \mathcal{L}_\perp$ functional is the central open problem for the weak sector. The correct order of magnitude — tens of GeV — emerges naturally from the medium parameters, without any separate gauge symmetry or Higgs mechanism. The Rayleigh-jet cap formula (§5.4) refines this to $m_W \approx 78.9\,\text{GeV}$, within 1.8% of the observed value. The Higgs sector, including the relation between the observed $m_H \approx 125\,\text{GeV}$ and the medium parameters, is treated separately in §5.3 below.

The Z boson and the Weinberg angle. The Z boson is the neutral reconnection channel, in which amplitude-mode and phase-mode caps mix. The SSV cap energy formula $m_{\text{cap}} = \pi R_{\text{cap}}^2 m_e c^2$ (with $P_0 = \xi = 1$) gives $m \propto R^2$, so the Standard Model tree-level relation $m_W = m_Z \cos\theta_W$ translates directly to

$$\frac{R_{\text{cap,W}}}{R_{\text{cap,Z}}} = \sqrt{\cos\theta_W}. \tag{33}$$

Using the PDG Weinberg angle as input, the tree-level Z cap radius is $R_{\text{cap,Z}} = R_{\text{cap,W}}/\sqrt{\cos\theta_W} \approx 236.8\,\xi$, giving $m_Z^{\text{tree}} \approx 90.0\,\text{GeV}$ (1.3% below the observed 91.2 GeV) — the same fractional deficit as for m_W. The Z-cap bending stiffness required by the equilibrium cubic (48) is $\lambda_{\text{bend,Z}}/\lambda_{\text{bend,W}} = 1.236$, consistent with the $\tau \to 0$ prediction $(R_{\text{cap,Z}}/R_{\text{cap,W}})^3 = 1/\cos^{3/2}\theta_W \approx 1.218$ at the 1.5% level.

The Weinberg angle itself remains an open gapbox: SSV does not yet predict θ_W from first principles. A scaling estimate at $R_{\text{cap,W}}$ gives $\tan^2\theta_W \sim \tau\alpha^2 R_{\text{cap,W}} \approx 0.20$, within a factor 1.5 of the observed value 0.30, consistent with a phase-to-amplitude stiffness ratio at the cap boundary. As a numerical remark, $\sin^2\theta_W \approx \phi/7 = 0.23115$ reproduces the PDG value 0.23122 to 0.03%; if this coincidence is not accidental, it would point to a golden-ratio quantisation of the SU(2)/U(1) mixing angle at the reconnection saddle.

5.3 The Higgs as Canvas: The Amplitude Mode of the Vacuum

In SSV the Higgs is not a separately postulated field. It is the amplitude mode of the same complex order parameter Ψ that produces every other particle in the framework. This section distinguishes two senses in which "the Higgs" appears in SSV, and locates the genuine hierarchy puzzle in the right place.

Two modes of the same medium. Linearising $\Psi = (\sqrt{\rho_0} + \delta\rho/2\sqrt{\rho_0})\, e^{i\theta_0 + i\delta\theta}$ around the saturated background gives two distinct small oscillations, exactly as in laboratory superfluids:

- **Goldstone mode** ($\delta\theta$): a phase oscillation at fixed density. In the literature-consistent reading discussed in §3.9, this is the leading candidate for the observed photon and propagates at the medium sound speed c.
- **Higgs mode** ($\delta\rho$): an oscillation of the condensate *density* itself. Massive, with gap $\Delta = c\sqrt{2m_0\lambda\rho_0}/\hbar$. This is the amplitude mode of the medium.

The two modes are governed by the same Lagrangian; the only distinction is which component of Ψ oscillates. The amplitude mode is gapped because compressing the saturated medium against the background pressure P_0 costs energy: the medium resists being anything other than what it is. This gap is a *structural* property of the LogSE Lagrangian — it exists at every point in the medium and is not associated with any localised event.

Defects vs. canvas. Every other particle in the framework is a *defect* in the medium: the electron is a toroidal hole, the proton a knotted vortex, the pion a vortex–antivortex link, the W a transient open cavity. All of these are absences or distortions of the saturated background. The amplitude mode is structurally different: it is the saturated background itself, set into oscillation. It is not a defect added to the canvas — it is the canvas, momentarily made visible.

Two readings of "the Higgs" in SSV. The word "Higgs" conflates two distinct things that must be kept separate:

(i) **The amplitude mode (structural).** The degree of freedom $\delta\rho$ is present everywhere in the medium; its gap $\Delta \simeq m_H c^2 \approx 125\,\mathrm{GeV}$ is a property of the Lagrangian parameters $\{b, \lambda, \rho_0\}$. This is not a particle; it is a channel of the medium, analogous to longitudinal sound in an ordinary fluid. The analogous amplitude mode in cold-atom condensates was first directly observed in 2012 [12].

(ii) **The 125 GeV denting event (derived).** The resonance observed at the LHC is a localised compression of $|\Psi|$ driven deep enough ($|\Psi| \to 0$ over a volume V^*) that the amplitude mode rings briefly before relaxing. Its energy is

$$E_H = P_0\, V^*, \tag{34}$$

where P_0 is the saturation pressure and V^* is the dent geometry set by the same 3D minimisation that determines the W end-cap volume V_{cap} (§5.4). The number 125 GeV is therefore a *derived* quantity calculable from $\{P_0, \alpha, \text{dent geometry}\}$, with no independent fundamental status.

Neither reading identifies m_H with a structural length scale of the medium's substrate. The quantity

$$\xi_{\mathrm{EW}} = \frac{\hbar}{m_H c} \approx 1.6 \times 10^{-18}\,\mathrm{m} \tag{35}$$

is the de Broglie wavelength of the amplitude-mode quantum; it is *not* the grain size of the medium. The medium's domain-of-validity scale is $\ell_{\mathrm{grain}} \equiv a_p \approx 2.10 \times 10^{-16}\,\mathrm{m}$ (Table 1), set by the heaviest stable defect, not by the Higgs denting event.

Yukawa coupling as canvas displacement. The Standard Model assigns each massive particle a Yukawa coupling $g_f = m_f/v$ to the Higgs field. In SSV this relation has a transparent mechanical meaning. An amplitude oscillation $\delta\rho$ couples to a defect in proportion to how much canvas the defect displaces from saturation; a heavier defect (larger ring, wider core, deeper density depression) interacts more strongly with $\delta\rho$. The proportionality $g \propto m$ is therefore not an empirical fact requiring a Yukawa sector — it is the geometric statement that bigger holes in the canvas couple more strongly to oscillations of the canvas itself.

The genuine hierarchy puzzle: the defect spectrum. The Standard Model hierarchy problem (why $m_H \ll m_{\mathrm{Planck}}$ given radiative corrections?) dissolves in SSV: there is no quadratically divergent radiative correction to m_H because ℓ_{grain}, not an external regulator, is the natural cutoff of the framework.

The *genuine* hierarchy question in SSV is different and sharper. The medium supports exactly two classes of stable defect: the toroidal vortex ring (electron, m_e) and the knotted breather (proton, m_p). Their spatial scales are

$$\text{electron ring radius:} \quad R^* = \xi/\alpha \approx 137\,\xi, \tag{36}$$

$$\text{proton core radius:} \quad a_p = \xi \cdot (m_e/m_p) \approx \xi/1836, \tag{37}$$

so the electron's major radius exceeds the proton's core by a factor of $137 \times 1836 \approx 2.5 \times 10^5$. Both structures arise from the same LogSE+chiral functional applied to different topologies. The framework does not yet predict this ratio from first principles: it requires the full 3D minimisation of the vortex-ring and knotted-breather energy functionals over $\mathcal{L} + \mathcal{L}_\perp$.

Structural composite mass ratio. The defect spectrum hierarchy of the previous paragraph is the SSV counterpart of the Standard Model Yukawa hierarchy: the ratio $m_p/m_e \approx 1836$ is determined by the relative scaling of two topology classes (the toroidal vortex ring and the knotted breather) under the same $\mathcal{L} + \mathcal{L}_\perp$ functional. Although the *individual* masses are at mixed status (Paper I tags m_e and $m_{\pi^\pm}$ as derived and m_p as candidate, issue #13), the framework already supplies one composite *mass-ratio* prediction whose structural form is fixed without further computation: the proton-to-pion mass ratio.

From the Paper I identifications

$$m_{\pi^\pm} \;=\; 2\mu_0, \qquad m_p \;=\; N_Y\, F\, \mu_0, \tag{38}$$

where $\mu_0 \equiv m_e/\alpha \approx 70\,\mathrm{MeV}$ is the electron-derived flux-tube mass scale, N_Y is the topological crossing factor of the trefoil Y-junction, and F is the 3D breather form factor, the ratio

$$\boxed{\frac{m_p}{m_{\pi^\pm}} \;=\; \frac{N_Y\, F}{2}} \tag{39}$$

follows immediately, with the empirical input μ_0 cancelling. At the current candidate-grade values $N_Y \approx 3.007$, $F \approx 4.47$,

$$\frac{m_p}{m_{\pi^\pm}} \;\approx\; 6.72, \tag{40}$$

within 0.1% of the observed ratio $m_p/m_{\pi^\pm} = 938.272/139.570 \approx 6.722$.

Cross-link to m_p/m_e. The same product $N_Y\, F$ also controls the proton–electron ratio through the definition $\mu_0 = m_e/\alpha$. Eliminating μ_0 from Eq. (38) gives the identity

$$2\,\frac{m_p}{m_{\pi^\pm}} \;=\; \alpha\,\frac{m_p}{m_e} \;=\; N_Y\, F. \tag{41}$$

This is a sharper statement than "the spectrum has a hierarchy". Three experimentally distinct dimensionless ratios ($m_p/m_{\pi^\pm}$, m_p/m_e, and α^{-1}) reduce to one structural quantity $N_Y\, F$ in the framework. Numerically, $2 \times 6.722 = 13.444$ and $(1/137.036) \times 1836.15 = 13.398$ — agreement at the 0.3% level, matching the present candidate-grade precision of N_Y and F. Eq. (41) is therefore an internal consistency *prediction*: any future first-principles computation of N_Y and F must reproduce both the 1836 and 6.72 ratios with the same single number.

Structural composite mass ratio

- **Structural form:** $m_p/m_{\pi^\pm} = N_Y F/2$ (Eq. (39)).
- **Numerical value:** 6.72 at present candidate-grade values, 0.1% below the observed 6.722.
- **Consistency identity:** $2(m_p/m_{\pi^\pm}) = \alpha(m_p/m_e) = N_Y F$ (Eq. (41)), reducing three observed dimensionless ratios to a single framework quantity at the 0.3% level.
- **Status:** structural — the form is fixed by the Paper I topology assignments (electron = toroidal vortex of radius ξ/α; pion = two-winding link; proton = trefoil Y-junction breather). The numerical values of N_Y and F inherit the candidate-grade status of the proton derivation (Paper I, issue #13). Closure of issue #13 promotes the ratio

from structural to derived.

Residual gap: the broader composite spectrum

The structural ratio above covers the lightest meson against the lightest baryon, plus the m_p/m_e consistency identity. Mass ratios within the larger spectrum — $m_K/m_{\pi^\pm}$ (kaon/pion), $m_\rho/m_{\pi^\pm}$ (vector meson/pion), m_Δ/m_p (delta resonance/nucleon), and the heavier-meson ladder — depend on α-harmonic placements (Paper I Table) that are themselves candidate-grade. A structural derivation for this larger sector requires identifying the specific surface-mode or Kelvin-wave excitation of the proton-trefoil breather and the pion-vortex link that each heavier state corresponds to. This is the natural continuation of §6 on the charged-lepton breathing-mode programme, applied to the baryon and meson sectors.

Connection to the W end-caps. The Higgs picture closes a loop in the weak-sector argument. The Rayleigh-jet estimate of the W mass (§5.4) depends on the cap term $E_{\text{caps}} \sim P_0 \cdot V_{\text{cap}}$, the cost of holding open a region where $|\Psi| \to 0$ against the saturation pressure P_0. This is the same P_0 that enters the Higgs denting energy (34): opening a W end-cap is driving the amplitude mode of the canvas to its full extent (collapse of the density to zero) over a smaller volume $V_{\text{cap}} < V^*$. The W and the Higgs are not two separate sectors with a coupling between them — they are two different denting events of the same medium, differing only in the geometry and depth of the cavity.

5.4 The W Boson as a Rayleigh Reconnection Jet

The estimate (32) treats the W boson as a point-like core-overlap event and misses the *geometry of the reconnection jet* — the dominant contribution to the mass. A classical analogue makes the structure explicit.

Physical picture: the Rayleigh jet. When a liquid droplet strikes a free surface it drives a cavity into the bulk; the cavity collapses axisymmetrically, focussing kinetic energy along the symmetry axis and ejecting a thin upward jet. The jet thins until its local radius reaches the capillary length, whereupon surface tension severs it and a new droplet pinches off. The energy stored at pinch-off is set by the *aspect ratio at the pinch point* — a purely geometric quantity independent of the initial cavity volume.

The SSV analogue maps precisely onto this process in three stages:

(i) **Cavity formation.** Two vortex defects approach within one ring radius $R^* \sim \xi/\alpha$. Quantum pressure suppresses $|\Psi|$ in the overlap region, opening a local cavity in the condensate — the amplitude channel activates.

(ii) **Jet formation.** The cavity collapses radially, driving an elongated vortex tube — an axially extended torus — along the reconnection axis. This tube carries exactly three polarisation degrees of freedom: two transverse oscillations of the tube wall, and one longitudinal mode from the stretching of the tube itself. The photon, a pure transverse phase-gradient wave, never activates the amplitude channel and therefore has no longitudinal mode and no end-caps.

(iii) **Pinch-off.** The tube thins until its minor radius reaches ξ, where quantum pressure dominates surface tension. The tube severs, depositing its energy into the outgoing reconfigured defects — the weak decay products.

The W boson is the *elongated vortex tube at the moment of pinch-off*: a transitional topology that exists only long enough to rewrite the identities of the interacting defects.

Energy of the jet at pinch-off. An elongated torus of length L^* and minor radius $a^* \sim \xi$ (the vortex *core* radius, not the ring radius R^*) has three energy contributions:

$$E_{\text{core}} \sim \rho_0 \, \xi^2 \, L^*, \tag{42}$$

$$E_{\text{tension}} \sim \rho_0 \kappa_0^2 \, L^*/a^{*2}, \tag{43}$$

$$E_{\text{caps}} \sim P_0 \, V_{\text{cap}}. \tag{44}$$

The first two terms are balanced by minimising over L^*, giving $L^* \sim \xi$ — the tube is roughly isotropic at pinch-off. The dominant contribution is the end-cap term, which has a qualitatively different physical origin from the other two: it is the cost of holding open a region of fully suppressed condensate ($|\Psi| \to 0$) against the enormous background saturation pressure $P_0 \gg \rho_0 c^2$. Every other energy in the problem is set by P_Q or the vortex tension $\rho_0 \kappa_0^2$; the cap cost is set by P_0 itself, making it the dominant term.

The scalar estimate (32) accounted only for $E_{\text{approach}} \sim P_Q \cdot a^3$ — the repulsive barrier the incoming defects must overcome to *initiate* reconnection. The cap cost $E_{\text{caps}} \sim P_0 \cdot V_{\text{cap}}$ is a separate, larger term that was not included. Since $P_0 \gg P_Q$, this naturally accounts for the missing factor of ~ 8.

The cap radius and the golden ratio. The cap volume $V_{\text{cap}} = \pi R_{\text{cap}}^2 \, \xi$ is set by the radius R_{cap} over which $|\Psi|$ is suppressed at each end of the W tube. Working backwards from the observed $m_W = 80.36\,\text{GeV}$ fixes $R_{\text{cap}} \approx 1.70\,\xi/\alpha$. This is tantalizingly close to

$$R_{\text{cap}} = \phi \, \frac{\xi}{\alpha}, \qquad \phi = \tfrac{1+\sqrt{5}}{2} \approx 1.618, \tag{45}$$

the golden ratio times the electron ring radius $R^* = \xi/\alpha$. Substituting (45) into E_{caps} with $P_0 \sim m_e/(2\xi^3)$:

$$m_W \approx \frac{\pi \phi^2 \, m_e}{\alpha^2} = \pi \phi^2 \times 9.60\,\text{GeV} \approx 78.9\,\text{GeV}, \tag{46}$$

within 1.8% of the observed value 80.36 GeV (PDG 2024).

Cap equilibrium and the chiral-shear bending stiffness. The cap is not a free surface: it is the boundary of a closed vortex ring held in mechanical equilibrium by three competing forces. Let R denote the cap radius. In SSV natural units ($P_0 = \xi = 1$) the total cap energy is

$$E(R) = \pi R^2 + 2\pi\tau R + \frac{2\pi\lambda_{\text{bend}}}{R}, \tag{47}$$

where πR^2 is the surface-pressure cost (area $\times P_0$), $2\pi\tau R$ is the vortex line-tension perimeter energy, and $2\pi\lambda_{\text{bend}}/R$ is the chiral-shear bending resistance that stiffens the ring against collapse. Setting $dE/dR = 0$ gives the equilibrium cubic

$$R^3 + \tau \, R^2 = \lambda_{\text{bend}}. \tag{48}$$

The vortex line tension is computed numerically from the planar LogSE profile (`vortex_cap_mass.py`): $\tau = 17.0\,\xi$ (core integral $+2\pi \ln(R_{\text{cap}}/r_{\max})$ tail). In the $\tau \to 0$ limit, setting $x = \alpha R/\xi$, equation (48) reduces to $x^3 = \alpha^3 \lambda_{\text{bend}}$. The unique positive solution $x = \phi$ (the golden ratio) requires

$$\lambda_{\text{bend}} = \frac{\phi^3}{\alpha^3} \, \xi^3 \approx 1.09 \times 10^7 \, \xi^3. \tag{49}$$

Evaluated at the physical $\tau = 17\,\xi$, the cubic places R_{eq} at ϕ/α to within the 7.7% line-tension correction (equation (48) with $R = \phi/\alpha$ gives $\lambda_{\text{bend}} = 1.17 \times 10^7 \, \xi^3$, vs the $\tau \to 0$ target $1.09 \times 10^7 \, \xi^3$). Thus $R_{\text{cap}} = \phi/\alpha$ is the unique equilibrium radius once $\lambda_{\text{bend}} = \phi^3/\alpha^3$ is established from the chiral-shear sector.

The SSV chiral-shear Lagrangian $\mathcal{L}_\perp = (\lambda_\perp/2)\int |\nabla \times \mathbf{j}|^2\, d^3r$ provides a natural coupling $\lambda_\perp = \alpha^{-2}$ (from $c_\perp = \alpha c$ at $k = 1/\xi$). Numerical integration of the core-bending correction to $\lambda_\perp$ over the planar vortex profile (`lperp_core_integral.py`) yields $\lambda_{\rm bend}^{\rm local} \approx 4.7 \times 10^4\, \xi^3$ — a factor 232 below the required value. The local curvature correction to $\mathcal{L}_\perp$ is therefore insufficient; the bending stiffness $\lambda_{\rm bend} = \phi^3/\alpha^3$ must arise from a *non-local* mechanism. The two leading candidates are: (a) quantisation of the chiral-shear energy of the Seifert surface of the trefoil (a non-perturbative topological contribution); and (b) a Casimir-like vacuum energy of the chiral mode between the cap boundary and the reconnection saddle.

Why the cap geometry is not directly accessible. The cap geometry is determined by the *dynamics* of the reconnection, not a static equilibrium, and cannot be observed directly: the event occurs at sub-ξ scales, lasts $\sim \xi/c \sim 10^{-25}$ s, and requires energies of order $P_0\xi^3$ to produce. The W mass itself is the only empirical constraint on $R_{\rm cap}$. Equation (46) holds as a conjecture: $R_{\rm cap} = \phi/\alpha$ follows analytically from the equilibrium cubic (48) once $\lambda_{\rm bend} = \phi^3/\alpha^3$ is established, but that identification requires deriving the non-local bending stiffness from the SSV Lagrangian — the remaining open step.

Why the W is massive and the photon is not. The photon lives entirely in the $\delta\theta$ channel and never suppresses $|\Psi|$ — it costs nothing against P_0. The W tube *requires* $|\Psi| \to 0$ at its end-caps: topology change cannot occur without locally dissolving the condensate, and the vacuum pressure P_0 must be held at bay for the duration of the event. Mass in SSV is fundamentally the cost of displacing the vacuum from itself — holding open a region where $|\Psi| < 1$ against P_0. The W end-caps are fully open wounds in the condensate, and P_0 is what closes them the instant the topology rewrite is complete. *The photon has no end-caps; the W boson has nothing but end-caps.*

5.5 Parity Violation

Parity violation in the weak interaction has a natural origin in the chirality of vortex reconnection. In a chiral superfluid (one whose ground state has a preferred handedness, which ours does — the toroidal electron vortex carries a fixed chirality tied to its charge), reconnections preferentially occur in configurations that preserve the global handedness. This selects left-handed neutrinos (defined hydrodynamically in §5.7 below as torsional pulses with a specific twist orientation) over right-handed ones, and produces the observed $(V - A)$ structure of the weak current without additional assumptions.

5.6 CP and the Strong CP Problem in a Medium That Remembers

The parity-violation argument of §5.5 relied on the medium having a preferred handedness. This raises the immediate question of whether the combined operation CP — charge conjugation paired with parity — restores the symmetry, and what the framework predicts for the small but observed CP violation in the kaon and B sectors. The answer requires confronting a distinction the Standard Model does not need to make: the difference between *symmetries of an equation* and *symmetries of a process.*

Two different things called "symmetry". In a quantum field theory built around a featureless vacuum, the two coincide: if the Lagrangian is invariant under a transformation T, then T maps every physical process onto another physical process governed by the same dynamics. SSV does not have a featureless vacuum. The medium is saturated, supports defects, and accumulates a record of every reconnection event in the form of its evolving topological tangle and the acoustic energy it has radiated (Paper III treats this explicitly as the origin of irreversible time). In such a medium, two distinct notions of "symmetry" must be carefully separated:

- **Formal symmetry of the equations.** A transformation T is a symmetry of the linearised $\mathcal{L} + \mathcal{L}_\perp$ functional if it leaves the equations of motion invariant. By this criterion, the dynamics of the saturated medium admits formal C, P, T, CP, CT, PT, and CPT operations, exactly as in the Standard Model.

- **Symmetry of physical processes.** A transformation is a symmetry of an actual physical process only if applying it produces another process consistent with the medium's accumulated state at the time the transformation is applied. By this stricter criterion, *no* discrete symmetry survives in SSV: every physical process deposits acoustic energy, leaves a chirality-marked imprint in the tangle, and modifies the medium that any putative symmetry partner would have to propagate through.

The Standard Model identifies these two notions because it works around a vacuum that does not record. SSV cannot. The medium remembers, and memory breaks every discrete symmetry at the level of physical processes, regardless of what the formal Lagrangian says.

Parity, charge conjugation, and CP, restated. In light of the distinction above:

- **P** is broken *at the Lagrangian level* by the chiral-shear coupling λ (§5.5). This is a strong form of breaking, comparable to the Standard Model's electroweak parity violation.

- **C** is broken at the Lagrangian level too, since the medium's chirality treats $n = +1$ and $n = -1$ vortices differently — the antivortex is not the same object as the vortex with the medium's handedness reversed.

- **CP** is a formal symmetry of the linearised dynamics (simultaneous winding and chirality flip), but *not* a symmetry of any physical process, because the process leaves a record that the CP-flipped partner would have to carry as well — and the flipped partner does not, because it propagates through a medium that has not accumulated the same wake.

The third bullet is the key shift from the Standard Model framing. CP violation in SSV is not a small departure from an underlying CP-exact dynamics, brought in by specific complex phases in the fermion mass matrix. It is the visible portion of an asymmetry that is structurally present in *every* process, made measurable only when reconnection geometry happens to amplify it past the threshold of detection.

Where the visible CP violation comes from. The kaon ε-parameter ($\sim 10^{-3}$) and the larger CP asymmetries in B-meson decays are not exotic phenomena requiring a special CP-violating sector. They are the systematic component of the much larger ensemble of CP-asymmetric reconnection events that occur continuously in the medium — the part that does not average to zero because the Y-junction packing geometry of the parent configuration biases one chirality of outcome over its CP-conjugate. The remainder averages out at experimental resolution and appears as "CP-symmetric noise" — but it is not truly absent, only unresolved. This is structurally identical to fluctuation–dissipation in classical statistical mechanics: the systematic asymmetry one measures (the analogue of dissipation) is the visible part of fluctuations that are asymmetric everywhere (the analogue of correlated noise).

The strong CP problem dissolves — but for a deeper reason. The Standard Model strong CP problem is the puzzle of why $|\theta_{\mathrm{QCD}}| \lesssim 10^{-10}$ when naturalness suggests $\theta \sim \mathcal{O}(1)$. The parameter θ_{QCD} measures the angle between the QCD vacuum and a CP-flipped reference vacuum; its smallness is mysterious because both vacua are equally available in the theory.

In SSV the strong sector is vacuum surface tension along flux-tube vortex segments (§4); there is no separate QCD vacuum. But this is not the deepest reason the problem dissolves.

The deepest reason is that a θ-angle measures deviation from a CP-symmetric reference state, and in a medium that records, no CP-symmetric reference state exists. The medium at any given moment carries a specific tangle history; the formally CP-flipped medium would carry a different history; there is no third state symmetric between them against which to measure θ. The Standard Model strong CP problem is well-posed only because the SM vacuum is featureless and admits unique CP-symmetric and CP-flipped versions. In SSV the question $\theta_{\mathrm{QCD}} =?$ is not answered as zero; it is dismissed as ill-defined.

The neutron electric dipole moment is therefore predicted to be zero at leading order in the same sense that the cosmic microwave background is isotropic at leading order: the leading-order calculation gives zero because the leading-order configuration is taken to be statistically isotropic, but the actual medium carries fluctuations and any specific measurement returns a small nonzero residue. The neutron EDM, like kaon CP violation, is a fluctuation-resolved asymmetry, not a tuning of a vacuum angle.

The matter–antimatter asymmetry as recorded chirality. Sakharov's three conditions for a baryon asymmetry — C and CP violation, baryon-number violation, and departure from thermal equilibrium — are all present in SSV, but the framework reads them differently than the Standard Model does. The medium's chirality is not a small CP-violating term added to a CP-symmetric dynamics; it is a property the medium has from its inception. Vortex reconnections do not accidentally violate baryon number; they are the mechanism by which topological charge changes. The early universe was not near thermal equilibrium that was disturbed by an out-of-equilibrium phase transition; it was the seeding event of the medium's record itself.

The observed baryon-to-photon ratio $n_B/n_\gamma \sim 6 \times 10^{-10}$ is therefore not a number to be derived from a small CP-violating phase in an otherwise symmetric theory. It is the surviving recorded chirality of the seeding event: the part of the initial asymmetry that crystallised into stable defects (matter) rather than annihilating against its partner (the near-equally-stable antimatter component). Why this number takes the value it does is open and properly belongs to Paper V's treatment of cosmogony, but the framework's reading of the question is sharp: the universe has matter because the medium remembers having been chirally seeded, and the small fraction n_B/n_γ measures how much of that initial chirality survived the early annihilation epoch.

Reconnection-memory derivation: the neutron EDM at leading order

The paragraphs above identified CP violation in SSV with the cumulative chirality bias of past reconnections — recorded in the medium and made visible when reconnection geometry amplifies it past detection threshold. This subsection develops the mechanism into a structural prediction for one CP observable: the neutron electric dipole moment. The derivation takes the neutron in its Paper I surface-locked composite form ($\mathcal{N} = \mathcal{T} \cup_{\Sigma_p} \mathcal{E}$, the proton trefoil $\mathcal{T}$ glued to a captured electron torus $\mathcal{E}$ at the breather skin Σ_p) and asks what permanent dipole moment the medium's chirality memory induces in this geometry at rest.

The chirality memory field. Define the local chirality memory density

$$\chi(\mathbf{x}, t) \;=\; \sum_{\text{events } i} \sigma_i \, K\big(\mathbf{x} - \mathbf{x}_i,\, t - t_i\big), \tag{50}$$

where $\sigma_i = \pm 1$ is the handedness of the i-th reconnection event (with respect to the medium's preferred chirality of §5.5) and K is the medium's memory kernel — a slowly decaying weight whose support is set by the propagation reach of the acoustic and chiral-shear modes emitted at the event. χ is a dimensionless scalar; its spatial average over any region is the surviving chirality fraction of all reconnection events whose memory still reaches that region. The cosmic average

$\langle\chi\rangle_{\rm cosmo}$ is the framework's reading of the baryon-to-photon ratio: $\langle\chi\rangle_{\rm cosmo} = n_B/n_\gamma \approx 6\times10^{-10}$ (§5.6, "matter–antimatter asymmetry as recorded chirality").

Coupling to the neutron lock. The captured electron torus $\mathcal{E}$ on Σ_p is held in equilibrium by the chiral-shear restoring potential $V_\perp = (\lambda_\perp/2)\,|\nabla\times\mathbf{j}|^2$ acting on the trefoil's surface current. A non-zero local χ couples to the same chiral-shear sector and biases the equilibrium position of $\mathcal{E}$ on Σ_p. To leading order in χ, the shift of the captured torus's centroid relative to the breather centre is

$$\delta\mathbf{r}_\mathcal{E} \;=\; \kappa_\chi\,\langle\chi\rangle_{R_n}\,R_n\,\hat{\mathbf{n}}, \tag{51}$$

where $R_n \approx 0.84\,\mathrm{fm}$ is the neutron radius, $\langle\chi\rangle_{R_n}$ is the average of χ over the neutron's interior volume, $\hat{\mathbf{n}}$ is the gradient direction of the chirality memory at the neutron, and κ_χ is a dimensionless coupling fixed by the chiral-shear Lagrangian. The lock stiffness $\partial^2V_\perp/\partial r^2$ at the surface-mode minimum places κ_χ at $\mathcal{O}(\alpha^n)$ for some $n \in \{1,2\}$ depending on whether the chiral-shear vertex enters linearly or at the one-loop self-energy of the surface mode. The numerical value of κ_χ requires the converged 3D minimisation of issue #15; the *structural form* of Eq. (51) does not.

Identifying $\langle\chi\rangle_{R_n}$. The neutron at rest in a quiet laboratory has no internal flavour oscillation channel; the relevant χ at its interior is dominated by the cosmic-background contribution, not by the kaon-scale fluctuations that drive ε_K. This is the structural distinction between the kaon and the neutron in this framework: the kaon's $K^0 \leftrightarrow \overline{K^0}$ oscillation continuously samples χ at the rate of the mixing, giving $\langle\chi\rangle_K \sim \varepsilon_K \approx 2.2\times10^{-3}$; the neutron has no such sampling channel, so $\langle\chi\rangle_{R_n}$ reduces to the cosmic background

$$\langle\chi\rangle_{R_n} \;\simeq\; \langle\chi\rangle_{\rm cosmo} \;=\; \frac{n_B}{n_\gamma} \;\approx\; 6\times10^{-10}. \tag{52}$$

The choice of the cosmic-background amplitude (rather than the kaon-scale amplitude) for the neutron is itself a structural prediction of the framework: only objects with an internal flavour-oscillation channel amplify χ to the ε-scale.

Leading-order prediction. Combining Eqs. (51) and (52),

$$\boxed{d_n \;=\; e\,|\delta\mathbf{r}_\mathcal{E}| \;=\; \kappa_\chi\,e\,R_n\,\frac{n_B}{n_\gamma}}. \tag{53}$$

Numerically, with $R_n \approx 0.84\,\mathrm{fm}$ and $n_B/n_\gamma \approx 6\times10^{-10}$,

$$d_n \;\approx\; \kappa_\chi \;\times\; 5\times10^{-23}\,e\!\cdot\!\mathrm{cm}. \tag{54}$$

At the conservative natural scale $\kappa_\chi \sim \alpha^2/\pi \approx 1.7\times10^{-5}$, $d_n \approx 9\times10^{-28}\,e\!\cdot\!\mathrm{cm}$, roughly one decade below the present experimental bound $|d_n| < 1.8\times10^{-26}\,e\!\cdot\!\mathrm{cm}$ [18] and within reach of the next-generation nEDM/n2EDM programmes targeting $\sim10^{-28}\,e\!\cdot\!\mathrm{cm}$. At the larger scale $\kappa_\chi \sim \alpha/\pi$ the prediction is $\sim10^{-25}\,e\!\cdot\!\mathrm{cm}$, which would already have been observed; the present null result therefore *requires* $\kappa_\chi \lesssim 4\times10^{-4}$ within the framework, narrowing the window for the deferred chiral-shear computation.

Consequences. The structural form $d_n \propto e\,R_n\,(n_B/n_\gamma)$ has three consequences:

- **Two observables, one coupling.** The neutron EDM and the baryon-to-photon ratio are predicted to be controlled by the same framework parameter κ_χ acting on the same chirality memory density χ. No separate CP-violating sector is invoked for either; the two distinct experimental numbers reduce to one in the framework.

- **Falsification.** Detection of $|d_n| \gtrsim 10^{-25}\, e\cdot$cm falsifies the framework as formulated — no natural κ_χ can be that large. A non-detection at the n2EDM sensitivity $\sim 10^{-28}\, e\cdot$cm would force $\kappa_\chi \lesssim 10^{-5}$, on the boundary of the natural α^2/π estimate.

- **Status.** Equations (53)–(54) are *structural*: the functional dependence on $(e,\, R_n,\, n_B/n_\gamma)$ is fixed by the framework; the dimensionless coupling κ_χ requires the converged 3D $\mathcal{L}+\mathcal{L}_\perp$ surface-mode calculation tracked under issue #15. This subsection therefore closes the "reconnection-memory to CP observables" gap (issue #35) for at least one observable; the residual gapbox below lists the observables still awaiting analogous derivations.

Residual gap: other CP observables

The reconnection-memory mechanism has now been applied to one CP observable (the neutron EDM, structural form fixed; numerical prefactor κ_χ deferred to issue #15). The remaining CP-sector observables are not addressed at this level:

- the CKM phase $\delta_{\text{CKM}} \approx 1.20\,\text{rad}$ as the cumulative chirality bias sampled by quark-flavour oscillations,
- the kaon ε-parameter and B-system CP asymmetries as the $\langle\chi\rangle$-amplified part of those oscillations' sampling rate,
- the baryon-to-photon ratio n_B/n_γ derived independently from the same memory kernel (rather than taken as input as in Eq. (52) above).

The structural framework for all three is the chirality memory density χ of Eq. (50); each requires identifying the specific oscillation channel that samples χ and computing the sampling rate against the same medium kernel K. These derivations are deferred to a sharper treatment of the chirality memory dynamics — the same $\mathcal{L} + \mathcal{L}_\perp$ minimisation that fixes κ_χ above is the natural starting point.

Forward connections. This subsection foreshadows two themes developed in later papers. The distinction between formal symmetries of equations and symmetries of physical processes is the thermodynamic content of Paper III: irreversibility is not something that emerges from a reversible substrate, it is the consequence of a medium that records, and the same logic that breaks T-symmetry at the process level breaks CP at the process level. The cosmological reading of the baryon asymmetry as recorded chirality of the seeding event connects to Paper V's treatment of cosmogony, where the universe's history is the medium's accumulated tangle and the question "what existed before?" becomes a question about the medium prior to its first records. The same whole-state coherence of the medium that breaks process-symmetries also underlies the holographic correlations of entangled particles: these correlations are properties of the medium's global saturated state, not signals propagating between measurement sites, and the experimental lower bounds on any "speed of spooky action" [14, 15] constrain hypothetical signal velocities and are vacuous in the present framework — they fall outside the propagation-speed analysis of §3 and §7 entirely.

5.7 Neutrinos as Torsional Radiation

Speculative subsection: motivated geometry, no derived spectrum

The treatment below identifies a candidate hydrodynamic object for the neutrino, but does not derive its mass, flavour structure, or oscillation phenomenology. Five open problems are listed at the end; none is addressed in this paper.

5.7.1 Geometric identification

A vortex reconnection is not silent. When two filaments exchange partners on the timescale $\tau \sim a/c$, the local twist density along each strand is discontinuously rearranged, and the excess torsion must be radiated away to satisfy conservation of angular momentum about the reconnection point. In a chiral superfluid the radiated mode is not a longitudinal phonon (that channel carries the photon and the gravitational signal) and not a transverse shear oscillation of a stable core (that channel carries the charged leptons): it is a *torsional pulse* — a propagating twist of the vortex-line phase field, unattached to any standing structure.

We tentatively identify the neutrino with this torsional radiation:

- **Charge**: zero, because the pulse carries no closed circulation loop — only a twist along an open propagation direction.
- **Helicity**: fixed by the chirality of the parent reconnection (see §5.5), giving the observed left-handed ν / right-handed $\bar{\nu}$ asymmetry without further assumption.
- **Range**: long, because a torsional pulse in the unsaturated longitudinal channel does not couple to the chiral-shear sector that gives charged leptons their rest mass — so it propagates at c to excellent approximation, with negligible interaction cross-section.
- **Production**: only in events that change vortex topology (reconnections) — i.e. in weak processes, never in EM or strong processes alone.

This identification is consistent with the qualitative picture sketched in Paper I (neutron decay as "straitjacket release"): the antineutrino is the torsional recoil pulse emitted when the compressed electron vortex unwraps from the proton breather.

5.7.2 What the framework does *not* yet predict

The torsional-pulse picture predicts a strictly massless, structureless radiation mode. This contradicts the established phenomenology in five distinct ways, each of which constitutes an open problem for the SSV programme:

(P1) **Non-zero mass.** Atmospheric and solar oscillation experiments require $\Delta m^2_{21} \approx 7.5 \times 10^{-5}\,\mathrm{eV}^2$ and $|\Delta m^2_{32}| \approx 2.5 \times 10^{-3}\,\mathrm{eV}^2$, so at least two mass eigenstates have $m_\nu \gtrsim 10^{-2}\,\mathrm{eV}$. A purely massless torsional pulse cannot accommodate this. A possible resolution is that the pulse is weakly self-localising in the chiral channel, acquiring a small effective mass from the same chiral-shear coupling λ that sets α — but no calculation currently exists.

(P2) **Three flavours.** The Standard Model has ν_e, ν_μ, ν_τ, paired with the three charged-lepton generations. In the present geometry a torsional pulse is characterised by its twist rate and its parent reconnection, neither of which has an obvious threefold structure. This is plausibly the same threefold-generation problem that affects the charged leptons (the muon and tau remain unidentified as specific breathing modes; see §6); a unified resolution of both would be desirable.

(P3) **Oscillations.** Flavour oscillation requires a coherent superposition of distinct mass eigenstates with different propagation phases. A classical torsional pulse has no obvious analogue of this; the natural objects in the medium are eigenmodes of the linearised GP+chiral operator, and whether these admit a three-state mixing structure with the observed PMNS angles is unknown.

(P4) **Dirac vs. Majorana.** The framework offers no current discriminator between $\nu = \bar{\nu}$ and $\nu \neq \bar{\nu}$. A torsional pulse and its parity-conjugate differ only in the sign of the twist; whether this constitutes a genuine particle/antiparticle distinction or a Majorana-like self-conjugacy depends on the global topology of the chiral medium and is not settled.

(P5) **Cross-section normalisation.** The Fermi constant $G_F = 1.166 \times 10^{-5}\,\mathrm{GeV}^{-2}$ should follow from the same reconnection-barrier calculation that fixes the W mass. Until that calculation is in hand (see §5), the predicted neutrino interaction cross-section is unconstrained.

The neutrino subsection is therefore presented as a geometric *hypothesis* in the same spirit as the neutron section of Paper I: a motivated identification that defers all quantitative claims to the full 3D reconnection computation. Problems (P1)–(P5) are flagged as deliverables for a subsequent paper in this series.

6 Charged Leptons: Higher Breather Modes

Status: muon partially placed, tau a ladder coincidence only

The electron is derived from first principles in Paper I. The muon is identified there with the lowest ring-breathing mode of the toroidal vortex, but the eigenfrequency itself is not yet computed — it is a placeholder filled by the ladder coincidence $m_\mu = (3/2)\mu_0$. The tau appears as a ladder rung at $25\frac{1}{2}\,\mu_0$ with no structural identification at all. Paper II adds nothing quantitative to either; this section serves only to locate the leptons within the present force-sector framework and to link the charged-lepton generation problem to the neutrino problem (P2) of §5.7.

6.1 The Muon: Ring-Breathing Mode of the Electron

The muon is identified in Paper I [1] (§Muon) with the lowest ring-breathing mode of the toroidal electron vortex — a periodic oscillation of the major radius R_e about its equilibrium value $R^* = \xi/\alpha$. The muon is therefore not a new defect type but the same toroidal vortex as the electron, set into a specific oscillatory mode by the chiral-shear coupling. Its production in weak processes is the same reconnection event that produces an electron, with the outgoing ring excited rather than ground-state.

The mass assignment $m_\mu = (3/2)\mu_0$ reproduces the observed muon mass to 0.61%. The open problem (Paper I, gapbox 2) is the first-principles computation of the ring-breathing eigenfrequency ω_{ring} in the 3D $\mathcal{L} + \mathcal{L}_\perp$ functional; the predicted frequency ratio $\omega_{\mathrm{ring}}/\omega_c = m_\mu/m_e \approx 206.77$ is the target. Paper II contributes no new computation here; the muon is mentioned for completeness, since otherwise the force-sector treatment would discuss neutrinos as the torsional companion of a charged lepton without ever introducing the charged lepton itself.

6.2 The Tau: Ladder Coincidence Without a Mode Assignment

The tau lands on rung $25\frac{1}{2}$ of the α-harmonic ladder ($m_\tau = 25.375\,\mu_0$, 0.49%) without any structural derivation. The ladder placement is consistent with the tau being a higher excited mode of the same toroidal topology — plausibly a second ring-breathing harmonic or a combined ring-breathing/chiral-shear mode — but no eigenmode calculation has been performed. In contrast to the muon, where the geometric picture (ring-breathing mode) is at least specified even if not yet computed, the tau identification is presently a numerology: a rung that fits the ladder, with no candidate hydrodynamic mode attached.

This matters for the present paper because the parity-violation argument of §5.5 and the neutrino identification of §5.7 both implicitly assume that each charged-lepton generation has a torsional companion (the corresponding neutrino). For the electron and muon this is empirically required and structurally consistent. For the tau the same correspondence is asserted but rests on the unresolved tau identification.

6.3 The Generation Problem

The Standard Model has three charged-lepton generations (e, μ, τ) paired with three neutrino flavours $(\nu_e, \nu_\mu, \nu_\tau)$. The framework presents a single unified version of this puzzle, since charged leptons and neutrinos in SSV are not independent objects but two facets of the same vortex configuration — the standing toroidal mode and the torsional pulse it emits during reconnection. A first-principles account of the threefold structure should therefore appear simultaneously in:

- **Charged leptons.** The eigenmode spectrum of the $\mathcal{L} + \mathcal{L}_\perp$ functional on the toroidal vortex background should yield the muon eigenfrequency (Paper I open problem 2) and, at the next harmonic, an identification of the tau. Whether the spectrum has exactly three accessible modes below the confinement scale — and why three — is unknown.
- **Neutrinos.** The same eigenmode structure should determine the flavour content of the radiated torsional pulse during reconnection (§5.7, P2 and P3), giving the three neutrino flavours and their oscillation phenomenology as a downstream consequence rather than as a separately tunable sector.

The threefold-generation problem is therefore one problem, not two. Its resolution is deferred to the same 3D minimisation that closes the W-mass gap (§5) and the α_G derivation (§7); all three open problems share the same $\mathcal{L} + \mathcal{L}_\perp$ computation as their target.

7 Gravity: Time-Delay Gradients from Interfering Pressure Waves

7.1 The Physical Picture

Gravity is the gradient of a time delay. Every massive defect is a breather (Paper I) — a localised region of Ψ oscillating at its Compton frequency — and it radiates a longitudinal pressure wave into the medium, with amplitude falling as $1/r$ from spherical dilution. A second breather, or a light wave, propagating through that radiated wave interferes with it; the time-averaged interference, through the medium's logarithmic nonlinearity, slows the local rate of change of Ψ. Since physical time is that rate of change (Paper III), the interference imposes a time delay, strongest near the mass and fading with distance. Its gradient bends every trajectory that crosses it toward the region of greatest delay — and that bending is gravity.

A single potential $\Phi(\mathbf{x})$ summarises the delay: the acceleration of free fall is $-\nabla\Phi$, and the fractional slowing of clocks is Φ/c^2. The present section establishes gravity as the fourth force-sector mode and fixes Newton's constant from the medium; the full development of the mechanism — the chain from interference to path bending, and the gravitational consequences that follow — is the subject of Paper IV.

7.2 The Proton as a Pulsating Breather

The source of the radiated wave is the breather itself. In Paper I the proton is a spherical breather mode — a periodically oscillating, localised density perturbation of the superfluid vacuum, sustained by its own knotted vortex skeleton and not by any external driving. Its eigenfrequency is the Compton frequency:

$$f_p = \frac{m_p c^2}{h} \approx 2.27 \times 10^{23}\,\text{Hz}, \tag{55}$$

with corresponding wavelength $\lambda_p = c/f_p \approx 1.3 \times 10^{-15}$ m. The breather radiates a longitudinal density wave into the medium at this frequency; the fractional density modulation $x(\mathbf{x}, t) \equiv \delta\rho/\rho_0$ falls as $1/r$ from spherical dilution. Every massive defect does the same, scaled by its mass-energy. Gravity is the interference of these radiated waves, and the next subsection computes it.

7.3 The Interference Cross-Term

The mechanism is contained in the LogSE. Where the radiated waves of several breathers overlap, the local density is $\rho/\rho_0 = 1 + \sum_i x_i$ with $x_i \equiv \delta\rho_i/\rho_0$, and the medium's logarithmic energy term expands as

$$\ln\Big(1 + \sum_i x_i\Big) = \sum_i x_i - \frac{1}{2}\sum_i x_i^2 - \sum_{i<j} x_i x_j + \cdots \tag{56}$$

The single-wave terms are each breather's own self-energy; the interaction between two breathers A and B is carried by the cross term

$$u_{AB} \;=\; -\,b\,\rho_0\,x_A\,x_B, \tag{57}$$

an *interference* term — the product of two waves. Three properties fix the character of gravity. Its time average $\langle u_{AB}\rangle = -b\rho_0\langle x_A x_B\rangle$ survives only for phase-correlated waves, so coherent breathers gravitate and incoherent disturbances do not. Its sign is fixed negative by the concavity of the logarithm: *gravity is attractive because the medium's nonlinearity is logarithmic.* And it is bilinear — with $x_A \propto m_A/r$ diluted to B's location and $x_B \propto m_B$ — so $\langle u_{AB}\rangle \propto -m_A m_B/r$, the Newtonian potential, with the inverse-square force its gradient. The coefficient is fixed by the breather's acoustic output in §7.5; the full development is given in Paper IV.

7.4 The Bjerknes Mechanism

This time-averaged interference of pulsating waves is, classically, the Bjerknes interaction [4]. For two bodies pulsating in a compressible medium with volume rates $\dot V_1$ and $\dot V_2$, Bjerknes obtained a time-averaged force

$$\langle F\rangle = -\,\rho_0\,\langle \dot V_1 \dot V_2\rangle\,\frac{1}{4\pi r^2}, \tag{58}$$

with r the separation. The cross-correlation $\langle \dot V_1 \dot V_2\rangle$ is exactly the interference term $\langle x_A x_B\rangle$ of §7.3 written in volume-velocity variables, and the geometric $1/(4\pi r^2)$ is the spherical dilution of a wave in three dimensions — not an independently postulated force law. We use Bjerknes' result and his volume-velocity form below to compute G; "force" is retained only as shorthand for the gradient of the interference. Its reinterpretation — why this is not a force across an inert gap, and why the breathers, as self-sustaining eigenmodes of Ψ, need no external driving — is developed in Paper IV.

7.5 Newton's Constant from the Medium

The quantitative interference calculation uses the breather's acoustic output to fix the coefficient of Φ left open in §7.3. It proceeds in four steps: the wave a breather radiates, the cross-term a second breather feels in it, the sub-grain suppression that makes that coupling weak, and the value of G that results.

Step 1: Acoustic wave of a pulsating breather. The proton pulsates at angular frequency $\omega_p = m_p c^2/\hbar$ with volume oscillation $V(t) = V_0 + \delta V_p \sin(\omega_p t)$, so $\dot V(t) = \omega_p \delta V_p \cos(\omega_p t)$. Both the static Bjerknes force and its radiative part are governed by the medium's sound speed c (§3.9). The standard Landau–Lifshitz result for a pulsating sphere [5] gives:

$$\delta P(r,t) = \frac{\rho_0}{4\pi r}\,\ddot V\Big(t - \frac{r}{c}\Big) = -\frac{\rho_0\,\omega_p^2\,\delta V_p}{4\pi r}\,\sin\Big(\omega_p t - \frac{\omega_p r}{c}\Big)\,. \tag{59}$$

Since the proton Compton wavelength $\lambda_p \approx 1.3\times 10^{-15}$ m is smaller than all macroscopic separations, gravitational interactions always occur in the radiation zone ($r \gg \lambda_p$). The time-averaged Bjerknes force nevertheless retains its $1/r^2$ form at all distances for in-phase oscillators, because the static component of the time-averaged force is independent of zone.

Step 2: The interference cross-term. A second identical proton at separation r is a resonant oscillator at ω_p, driven in phase by the same medium. The time-averaged cross-term of §7.3, in volume-velocity variables, is the average over one cycle:

$$\langle \mathbf{F}_{12} \rangle = -\frac{\rho_0}{4\pi r^2} \langle \dot{V}_1(t)\, \dot{V}_2(t) \rangle\, \hat{r}, \tag{60}$$

with

$$\langle \dot{V}_1 \dot{V}_2 \rangle = \tfrac{1}{2}\omega_p^2 (\delta V_p)^2 > 0, \tag{61}$$

giving an attractive force $\propto 1/r^2$. For two objects of different masses the pulsation amplitude scales as $\delta V \propto m/\rho_0$ (Step 3), so the force is proportional to $m_1 m_2/r^2$ — Newton's law.

Step 3: Pulsation amplitude and sub-grain suppression. The fractional volume excursion of a breather is set by the ratio of its rest-mass energy to the medium's compressive energy over the breather volume:

$$\varepsilon_p \equiv \frac{\delta V_p}{V_0} = \frac{m_p c^2}{\rho_0 c^2 \cdot V_0} = \frac{m_p}{\rho_0 V_0}, \tag{62}$$

where $P_0 \approx \rho_0 c^2$ and $V_0 = \frac{4}{3}\pi a_p^3$ with $a_p = \hbar/(m_p c)$ the proton Compton radius. Since $\delta V \propto m$, the Bjerknes force scales as $m_1 m_2$ as required.

The proton is a *sub-grain* oscillator: its Compton radius a_p is smaller than the medium healing length $\xi = \hbar/(m_e c)$ by the mass ratio

$$\frac{a_p}{\xi} = \frac{m_e}{m_p} \approx \frac{1}{1836}. \tag{63}$$

The medium cannot resolve structure below ξ; consequently the proton couples to the long-range density wave with a suppressed *acoustic monopole moment* Q_p. For a sub-grain spherical oscillator, coupling to wavelengths $\lambda \gg \xi$ is suppressed by the ratio of source volume to grain volume:

$$Q_p \sim \delta V_p \cdot \left(\frac{a_p}{\xi}\right)^3 = \delta V_p \cdot \left(\frac{m_e}{m_p}\right)^3. \tag{64}$$

The coupling hierarchy across the three stable objects is:

$$\eta_\gamma = 1 \;\gg\; \eta_e \sim \alpha \;\gg\; \eta_p \sim \left(\frac{m_e}{m_p}\right)^3, \tag{65}$$

where the photon couples with unit efficiency (it *is* the medium wave), the electron couples through its toroidal topology (suppressed by α), and the proton couples as a sub-grain monopole (suppressed by $(m_e/m_p)^3$). Gravity is weak not because of a separate fine-tuning, but because the proton is small relative to the medium grain.

Step 4: Newton's constant from the medium. Substituting Q_p into (60) and matching to $\langle F_{12} \rangle = G m_p^2/r^2$:

$$G = \frac{\rho_0\, \omega_p^2\, Q_p^2}{8\pi\, m_p^2}. \tag{66}$$

A full evaluation of Q_p requires the 3D proton breather profile and is deferred to a companion paper. A structural consistency check uses the gravitational coupling constant $\alpha_G \equiv G m_p^2/(\hbar c)$ and the proton mass from the Paper I ladder, $m_p = N_p\, m_e/\alpha$ with $N_p \approx 13.44$:

$$G = \frac{\alpha_G\, \hbar c\, \alpha^2}{N_p^2\, m_e^2}. \tag{67}$$

This is a structural consistency check: α_G is taken from CODATA rather than derived from first principles, confirming that the Paper I mass ladder is internally coherent with the gravitational sector. Inserting $\alpha_G = 5.906 \times 10^{-39}$ (CODATA) and $N_p = 13.44$:

$$G_{\text{pred}} = \frac{5.906 \times 10^{-39}\, \hbar c\, \alpha^2}{(13.44)^2\, m_e^2} = 6.634 \times 10^{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2}, \tag{68}$$

within 0.6% of $G_{\text{obs}} = 6.674 \times 10^{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2}$. The residual is consistent with the imprecision in N_p (form factor $F \approx 4.47$ is numerically determined; see Paper I).

Open problem: first-principles derivation of α_G

G is not an independent constant of the framework. Equation (67) expresses it in terms of $\{\hbar, c, \alpha, m_e, \alpha_G\}$, where the first four are fixed in Paper I. The remaining open problem is computing the sub-grain suppression factor $(a_p/\xi)^n$ from the 3D proton breather profile, which would yield α_G without empirical input. The 0.6% structural agreement is evidence of framework consistency, not a completed derivation.

Gravitational and electromagnetic radiation share the Goldstone channel. In the literature-consistent reading of the medium (§3.9; Volovik [16], Zloshchastiev [17]), electromagnetic and gravitational radiation are both phase-channel ($\delta\theta$) disturbances of Ψ, propagating at the medium's sound speed c. They differ only in their multipole-source structure: electromagnetic radiation arises from the chirality dipole of an accelerating toroidal vortex, while gravitational radiation arises from the mass-energy quadrupole of an accelerating pulsating breather. Because they share the carrier channel, both propagate at exactly c and arrive simultaneously from any astrophysical source.

This is consistent with the GW170817 observation. The neutron-star merger GW170817 produced gravitational and electromagnetic signals separated by $\sim 1.7\,\text{s}$. In SSV this delay is attributed entirely to source physics — the relativistic-jet launch, cocoon formation, and jet breakout that delay the gamma-ray burst relative to the moment of merger — and carries no implication for any difference in propagation speed. This is in full agreement with the mainstream astrophysical interpretation [13]. The chiral-shear mode at $c_\perp$ is an internal mode of the medium that mediates the static near-field Coulomb interaction between vortices (§3.9); it is not the carrier of any astrophysical radiation.

7.6 Scope: the Full Treatment in Paper IV

The present section has established gravity as the fourth force-sector mode — the interference of breather pressure waves, building a time-delay gradient that bends trajectories — and has fixed Newton's constant from the medium. The conceptual development of the mechanism is carried out in Paper IV: the formal account of update capacity, the four gravitational consequences (free fall, time dilation, light deflection, redshift), the universality of the coupling, the absence of a graviton, the bridge to emergent geometry (Paper VII-b), and the connection to the recording medium of Paper III.

8 Unified Picture: One Medium, Four Modes

We can now survey the four forces from a single vantage point. Table 2 summarises the correspondence and adds the Higgs as the one remaining mode of the medium that is not an interaction between defects.

Several structural features of this table deserve comment.

Table 2: The four forces and the Higgs mode as facets of the saturated superfluid vacuum. The four forces describe how defects in the medium interact; the Higgs row is structurally different — it is the medium itself oscillating, not an interaction between defects.

Sector	**Mode type**	**Source**	**Carrier**	**Range**
Electromagnetism	Transverse flow	Chirality ($\pm$)	Transverse phonon	∞
Strong	Surface tension	Topological charge	Phase oscillation	$\sim a/\alpha$
Weak	Reconnection	Core overlap	Amplitude mode (W, Z)	$\sim \xi$
Gravity	Pressure-wave interference	Mass-energy (breather)	Longitudinal density wave	∞
Higgs	Amplitude mode	The medium itself	$\delta\rho$ oscillation	$\sim \xi_{\mathrm{EW}}$

Range. The infinite-range forces (EM and gravity) are both mediated by massless phonon modes — transverse and longitudinal, respectively. The short-range forces involve massive excitations: the gluon acquires an effective mass from the string tension at distances exceeding a/α, and the W/Z are the high-energy amplitude modes of the core. The hierarchy of ranges $r_W \ll r_{\mathrm{strong}} \ll r_{\mathrm{EM}}, r_{\mathrm{gravity}}$ follows directly from the hierarchy $\xi \ll a/\alpha \ll \infty$.

Coupling strengths. All couplings are fixed by the same two numbers: α and $\mu_0 = m_e/\alpha$. The strong coupling is ~ 1 (order-unity surface tension, no small factor), the electromagnetic coupling is $\alpha \approx 1/137$ (transverse-to-longitudinal wave speed ratio), the weak coupling is α^2 (two powers of the core penetration suppression), and the gravitational coupling is $Gm_p^2/\hbar c \approx 6 \times 10^{-39}$ (set by the tiny ratio of the proton's acoustic cross-section to c^4/G, which encodes the vast hierarchy between P_0 and $m_p c^2$).

Universality of gravity. Every other force is selective: EM acts on chirality, strong on topological charge, weak on core overlap. Gravity (Bjerknes) acts on *acoustic cross-section*, which is determined by mass-energy alone. This is the mechanical root of the equivalence principle.

Defects vs. canvas. The four force rows of Table 2 share a common pattern: each describes how some property of a defect (chirality, topological charge, core overlap, mass-energy) sources a pressure response of the medium that another defect feels. The Higgs row breaks this pattern. Its source is not a defect but the medium itself; its carrier is the same density mode that, when sourced by a moving breather, transmits gravity — but intrinsically rather than as an interaction. A reader who finds the Higgs "out of place" in a force-unification table is correctly identifying the asymmetry: the Higgs is not a fifth force, it is the medium's own amplitude mode made briefly visible. Its inclusion makes explicit that the same Ψ-field substrate produces both the four-force interaction structure and the matter / canvas distinction that underlies it.

Particle content on the same substrate. The same medium produces the full matter content of the Standard Model: electrons as toroidal vortex rings, baryons as knotted vortex breathers, charged leptons as higher breathing harmonics (§6), neutrinos as torsional radiation from reconnection (§5.7), and the discrete symmetry structure (P-violation, the absence of exact CP, the absence of a strong CP problem) as direct consequences of the medium's chirality and recorded history (§5.6). Particles, forces, and symmetries are not three separate sectors with mutual couplings; they are three different ways of asking questions about the same hydrodynamic substrate.

No new constants. The framework introduces no coupling constants beyond those already fixed by the mass spectrum of Paper I. The four forces, the Higgs sector, the particle content, and the discrete-symmetry structure share the same parameters $\{c, \alpha, \mu_0\}$; they are not independent sectors with mutual couplings, but four different windows into the same hydrodynamic pressure field.

9 Testable Predictions

The framework makes a number of predictions that go beyond what is currently explained within the Standard Model. We organise them by sector to make clear which experiments would test which structural claims.

Force-sector and gravitational predictions

1. **Graviton spin.** The longitudinal acoustic phonon of the superfluid is naively spin-0. We predict that the graviton in this framework has an *effective* spin-2 character that emerges from the transverse-traceless projector in the linearised acoustic metric — a testable statement about the polarisation modes of gravitational waves. The ratio of breathing-mode to quadrupolar-mode amplitude in binary inspirals should carry a small superfluid correction at order v/c in the post-Newtonian expansion.

2. **Gravitational Casimir correction.** The Bjerknes force between a proton and any other mass includes a next-to-leading-order correction from the *quantum pressure* term, analogous to the Casimir effect. At sub-micron separations, this produces a deviation from the $1/r^2$ law of order $(\delta r/r)^2 \sim (a/r)^2$, where $a \sim 10^{-15}$ m is the vortex core size. This is below current precision for laboratory experiments but could in principle be detected at hadron-scale separations in neutron scattering experiments.

3. **Gravitational-wave memory.** The gravitational-wave memory effect (a permanent shift in detector test-mass separation following the passage of a strong GW signal) is predicted to be present at the Christodoulou level — not as a peculiar property of asymptotic spacetimes but as a direct consequence of a medium that records (Paper IV). The framework sets the same magnitude as conventional GR; the structural claim is that memory is the natural state of affairs in SSV, not an exotic signal. Distinguishing SSV from GR on this point requires precision detection of the fluctuation spectrum around the memory mean, which is beyond current sensitivity.

4. ***W*-mass geometric fit awaiting derivation.** Our scalar estimate $E_{\text{recon}} \sim m_e/\alpha^2 \approx 9.6\,\text{GeV}$ is a factor of ~ 8 below the observed W mass. The Rayleigh-jet cap formula (§5.4, equation (45)) gives $m_W \approx 78.9\,\text{GeV}$ — within 1.8% of the observed value. The cap equilibrium analysis (§5.4) shows that $R_{\text{cap}} = \phi/\alpha$ follows analytically from the chiral-shear bending stiffness $\lambda_{\text{bend}} = \phi^3/\alpha^3$ via the equilibrium cubic $R^3 + \tau R^2 = \lambda_{\text{bend}}$. The local $\mathcal{L}_\perp$ curvature correction falls short by a factor 232; the remaining open step is deriving the non-local bending stiffness ϕ^3/α^3 from the topological structure of the cap (Seifert surface of the trefoil, or chiral-mode Casimir energy at the reconnection saddle).

5. **Running of α at high energy.** The chiral-shear coupling α is a medium property, not a bare constant. At momentum scales approaching $\mu_0 = m_e/\alpha$, the chiral-shear sector begins to couple to the longitudinal modes through the nonlinear term in the Gross–Pitaevskii equation. This produces a logarithmic running of α that should match the QED renormalisation group equation — a self-consistency check that has not yet been completed in detail.

Particle-content predictions

6. **Higgs as condensed-matter analogue.** Since the Higgs is identified as the amplitude mode of the same order parameter that carries every other particle (§5.3), its phenomenology should match that of the amplitude mode in cold-atom condensates near saturation [12], scaled up by the medium parameters. The decay channels of the Higgs reflect the eigenmode structure of the medium's amplitude oscillations; small deviations from Standard Model branching ratios at high precision (above current LHC sensitivity) would be expected if the amplitude mode is genuinely a hydrodynamic excitation rather than a fundamental scalar. The framework does not predict a specific deviation magnitude; the structural claim is that the Higgs is not a separately tunable sector.

7. **No fourth-generation lepton.** The framework predicts that the toroidal vortex background supports a finite eigenmode spectrum; the muon and tau are placeholders for the lowest two ring-breathing harmonics (§6). Whether the spectrum has exactly three accessible modes below the confinement scale is open, but the threefold Standard Model generation structure — if it follows from this eigenmode structure — would imply that searches for a fourth charged lepton or a fourth neutrino flavour should return null at all accessible energies. Any positive result would falsify the eigenmode interpretation of the lepton generations.

8. **Neutrino emission only in topology-changing events.** In the SSV picture, neutrinos are torsional pulses emitted only when vortex topology is rewritten — i.e. in weak-sector reconnections. No neutrino emission accompanies any purely electromagnetic or strong-sector process, including high-energy QED vertices, hadronic resonance decays that proceed through the strong channel, or coherent photon scattering, regardless of available phase space. This is a sharper statement than the Standard Model selection rule (which is a flavour-current statement); here it is a structural impossibility, since no torsional pulse can be sourced without a reconnection. Any observation of neutrino emission unaccompanied by topology change in the source would falsify the framework.

9. **No magnetic monopoles.** The magnetic field is defined as $\mathbf{B} \equiv c_\perp^{-1}\nabla\times\mathbf{v}_\perp$ (Section 3.6), so $\nabla\cdot\mathbf{B} \equiv 0$ is a vector identity, not a postulate. A magnetic monopole would require a point-like topological defect sourcing the transverse velocity as a scalar potential — but the GPE supports only vortex-line defects (which source the *rotational* part of $\mathbf{v}_\perp$, carrying electric charge). The absence of magnetic monopoles is a topological theorem of the framework: no object with the required topology can exist in a superfluid order parameter in three spatial dimensions. All current and future monopole searches should return null results.

Discrete-symmetry and CP-sector predictions

10. **Neutron electric dipole moment from reconnection memory.** The strong CP problem dissolves in SSV (§5.6): there is no CP-symmetric reference vacuum, so θ_{QCD} is not a free parameter to be tuned. The reconnection-memory derivation of §5.6 gives the leading-order structural form

$$d_n \;=\; \kappa_\chi\, e\, R_n\,\frac{n_B}{n_\gamma} \;\approx\; \kappa_\chi \;\times\; 5\times10^{-23}\, e{\cdot}\text{cm},$$

with κ_χ a dimensionless chiral-shear coupling deferred to issue #15. At the natural scale $\kappa_\chi \sim \alpha^2/\pi$ this gives $d_n \sim 10^{-27}\, e\cdot\text{cm}$, one decade below the present bound $|d_n| < 1.8\times10^{-26}\, e\cdot\text{cm}$ [18] and within reach of nEDM/n2EDM at $\sim 10^{-28}\, e\cdot\text{cm}$. The same chirality memory density χ underlies both this prediction and the cosmological baryon asymmetry, tying two distinct CP observables to a single framework coupling.

11. **No axion required.** The Peccei-Quinn mechanism and its associated axion are introduced in the Standard Model to explain the smallness of θ_{QCD}. Since SSV does not have a θ_{QCD} parameter to begin with, no axion is required, predicted, or expected. Detection of an axion at any mass would not falsify SSV directly — axions could exist as additional medium excitations not yet identified in the framework — but their detection would remove the SSV's most distinctive structural prediction in the strong-CP sector.

12. **Baryon asymmetry from chiral seeding.**

    The observed baryon-to-photon ratio $n_B/n_\gamma \sim 6 \times 10^{-10}$ is interpreted as the surviving recorded chirality of the seeding event (§5.6). Quantitatively, the prediction is that this ratio should follow from the same chirality bias that determines the CKM phase — i.e. from a single number measured in two completely different experiments. This is a sharp consistency prediction: whatever calculation eventually produces the CKM phase from reconnection geometry should also produce the baryon-to-photon ratio. No separate cosmological CP-violating sector should be required.

Falsification structure

The above predictions are not equally falsifiable, and it is worth being explicit. The structural predictions (no monopoles, no axion required, neutrinos only from topology change, no fourth-generation leptons) admit clean falsification: a single confirmed positive observation in any of these categories would invalidate the framework as currently formulated. The quantitative predictions (graviton spin corrections, W mass, Higgs phenomenology, n_B/n_γ) require performing the 3D $\mathcal{L} + \mathcal{L}_\perp$ minimisation before they can be compared against observation; their failure mode is not a single-experiment refutation but the inability to reproduce known numbers from the geometry. The gravitational-wave memory and gravitational Casimir predictions are quantitatively close to GR/Standard-Model expectations and would require precision improvements beyond current capability to discriminate.

10 Bridge to Thermodynamics and Irreversible Time

The same superfluid order parameter Ψ that underlies the four forces also governs irreversible thermodynamics. One further connection to a subsequent paper is worth flagging here, beyond the forward-pointer already given in §5.6 (CP and memory).

Time and entropy (Paper III). The two flagged places where the present paper meets the irreversibility framework — the symmetry/process distinction in CP, and the time-averaging step in Bjerknes gravity — are not isolated remarks but the visible edges of a single substrate question: in a medium whose state at any moment is the record of all reconnections that preceded it, what counts as an instantaneous quantity? Paper III treats this question directly: the arrow of time is identified with the growth of the topological tangle, and the second law is the statement that the medium accumulates record monotonically. The cosmological side of the same picture — the CMB thermal spectrum read as acoustic noise integrated from all reconnection events since the seeding — is the subject of a separate future paper (Paper IX, on the primordial phonon bath), not of the present paper. The force-sector mean-field treatment of the present paper is the static envelope of that fuller dynamics.

11 Discussion and Open Problems

The framework presented here is mechanically complete for the force sector: each of the four interactions has a definite hydrodynamic origin, and the extensions added in this paper — the

Higgs as the canvas's amplitude mode, neutrinos as torsional radiation, charged leptons as higher breather harmonics, and CP violation as a fluctuation residue in a medium that records — fill out the particle content and the discrete-symmetry structure of the Standard Model on the same underlying hydrodynamic substrate. No additional fields, gauge groups, or coupling constants are introduced beyond those fixed by the mass spectrum of Paper I.

What remains is computation, not conceptual machinery. The open problems collected here divide into two groups: a single core calculation on which most of them depend, and a small number of structural questions that are genuinely independent.

The core open computation

The most significant feature of the present framework's open-problem landscape is that it collapses several apparently independent puzzles of the Standard Model into a single deferred computation: the full 3D time-dependent minimisation of the $\mathcal{L} + \mathcal{L}_\perp$ functional during a vortex reconnection event. The following observables are all determined by the geometry of that minimisation, and none can be computed in isolation from the others:

1. **The W boson mass.** The scalar estimate $E_{\text{recon}} \sim m_e/\alpha^2 \approx 9.6\,\text{GeV}$ accounts only for the quantum-pressure barrier that initiates reconnection. The dominant cost is $E_{\text{caps}} \sim P_0 \cdot \pi R_{\text{cap}}^2 \xi$ — holding open the two fully-suppressed end-caps against the saturation pressure P_0. The formula $R_{\text{cap}} = \phi\,\xi/\alpha$ (equation (45)) gives $m_W \approx \pi\phi^2 m_e/\alpha^2 \approx 78.9\,\text{GeV}$ — within 1.8% of the observed value. The cap equilibrium analysis (§5.4) shows this is not a bare fit: $R_{\text{cap}} = \phi/\alpha$ is the unique equilibrium of the cubic $R^3 + \tau R^2 = \lambda_{\text{bend}}$ (line tension $\tau = 17\,\xi$, computed) when $\lambda_{\text{bend}} = \phi^3/\alpha^3$. The local $\mathcal{L}_\perp$ correction falls short by 232×; the target is now to derive $\lambda_{\text{bend}} = \phi^3/\alpha^3$ from the non-local cap topology.

2. **The neutrino sector (P1–P5 of §5.7).** Mass, three-flavour structure, oscillation phenomenology, Dirac/Majorana character, and the Fermi constant G_F all follow from the same reconnection geometry: the torsional pulses radiated during the event carry whatever flavour structure, mass spectrum, and mixing the linearised GP+chiral operator supports as eigenmodes around the 3D reconnection background.

3. **The charged-lepton generations (§6).** The muon eigenfrequency (Paper I open problem 2) and the tau identification (currently a bare ladder coincidence) are eigenmodes of the same operator, evaluated on the toroidal vortex background rather than the reconnection background. Paper I and Paper II treat them separately for expository reasons; they are mathematically the same problem.

4. **CP violation observables (§5.6).** The CKM phase, the kaon ε-parameter, the B-system CP asymmetries, the neutron electric dipole moment, and the baryon-to-photon ratio $n_B/n_\gamma \sim 6 \times 10^{-10}$ are all consequences of the chirality bias in reconnection outcomes. In a medium that records, these are not small departures from a CP-symmetric reference dynamics; they are the visible component of asymmetries present in every event, made measurable when reconnection geometry happens to amplify them past detection threshold.

5. **The strong sector at two loops.** Identification of colour charge with phase balance at Y-junctions, and the derivation of the QCD β-function from quantum-pressure softening, should be carried to two-loop order using the same 3D reconnection treatment. Comparison with precision QCD measurements provides an independent calibration of the reconnection geometry.

These are not five independent open problems but five facets of the same computation. Each of the corresponding Standard-Model parameters (the electroweak scale, the PMNS matrix, the lepton mass ratios, the CKM phase, $\Lambda_{\rm QCD}$) is currently fitted to data; the framework's claim is that all are determined by the geometry of one event, computable in principle without empirical input. This is a sharp, falsifiable claim — the geometry will produce specific numbers that either match observation or do not.

Structurally separate open problems

Three further open problems stand outside the core computation, in the sense that their resolution requires different tools.

6. **Newton's constant: α_G from the proton breather.** The structural formula (67) reproduces $G_{\rm obs}$ to 0.6% using the Paper I mass ladder. The remaining task is computing the sub-grain acoustic monopole suppression factor $(a_p/\xi)^n$ from the 3D *static* proton breather profile, which would yield $\alpha_G = Gm_p^2/(\hbar c) \approx 5.906 \times 10^{-39}$ without empirical input. This is a different computation from the reconnection one (a static eigenmode rather than a dynamical event), but uses the same $\mathcal{L} + \mathcal{L}_\perp$ functional.

7. **Spin-2 graviton from linearised Madelung.** A systematic post-Newtonian expansion of the Madelung equations around a pulsating source is needed to confirm the spin-2 angular pattern of gravitational radiation. This is a perturbative calculation around the static gravity picture of §7, independent of the reconnection geometry.

8. **The canvas-grain hierarchy (§5.3).** The Higgs mass corresponds to a canvas grain length $\xi_{\rm EW} \approx 1.6 \times 10^{-18}$ m, roughly 10^5 times finer than the vortex healing length ξ set by m_e. The framework does not solve the hierarchy problem; it restates it as the geometric question of why the medium has two natural length scales separated by this factor. Note that the Standard Model fine-tuning instability does *not* transfer: the cutoff ξ is itself a property of the medium, so there are no quadratically divergent radiative corrections that would drag m_H to a higher scale. The problem is structural, not a fine-tuning.

Boundary with Paper III

Two specific places in this paper depend on operations whose finer-resolution treatment is the subject of Paper III on irreversible thermodynamics. They are flagged here for completeness; neither obstructs the mean-field claims of the present paper.

9. **Time-averaging in a recording medium (Paper IV).** The Bjerknes time-averaging that produces Newton's law is a classical operation; in a medium that accumulates a record of every reconnection event, the corrections to the leading-order static result carry information about pulsation history, and reproduce the gravitational-wave memory effect (Christodoulou) as a natural consequence.

10. **Discrete symmetries at the level of physical processes (§5.6).** The distinction between formal symmetries of the $\mathcal{L} + \mathcal{L}_\perp$ functional and symmetries of physical processes — which are broken by the medium's accumulated record — is the conceptual root of irreversibility. Paper III makes this distinction the foundation of its treatment of time and entropy.

What is closed

It is worth stating explicitly what the present paper closes, since the length of the open-problems list could otherwise mislead. The mean-field mechanical content of all four forces is fixed: each is a calculable pressure response of the saturated medium, with coupling constants determined by $\{c, \alpha, \mu_0\}$ from Paper I. The Higgs is identified with the amplitude mode of the same order parameter Ψ, with no separately postulated field. Magnetic monopoles are topologically forbidden, not merely absent. The ultraviolet divergence of perturbative quantum gravity does not arise. The strong CP problem and the cosmological constant fine-tuning problem do not arise. Parity violation, the $(V - A)$ structure of the weak current, the equivalence principle, the universality of gravitational coupling, and the impossibility of magnetic monopoles all follow from the single hydrodynamic substrate without additional assumption.

The remaining open problems are computational, not foundational. Each has a definite formulation within the framework already established here and in Paper I, and a clear path to numerical resolution. The framework's most distinctive structural feature is the unification of what the Standard Model treats as independent puzzles — the W mass, the neutrino sector, the charged-lepton generations, and the CP-violation observables — into facets of a single 3D minimisation. Whether that minimisation, when finally performed, produces the right numbers is the empirical test the framework sets for itself.

12 Conclusion

We have proposed that the four fundamental forces emerge from a single superfluid vacuum as distinct mechanical modes. The present paper is a programmatic and structural report; the precise epistemic status of each sector is summarised below.

What the paper establishes. The mechanical origin of the electromagnetic near-field interaction — chiral flow gradients around toroidal vortices, with Coulomb's law and the Biot–Savart force following from the Bernoulli pressure field — is reached at the level of structural mechanism using the chiral-shear coupling α and the medium parameters fixed in Paper I. The propagation speed of light in the provisional radiative identification is tied to the chiral-shear mode speed $c_\perp$ rather than the medium's sound speed c; this is a known open issue (§3.9) whose resolution requires re-deriving the photon as the Goldstone mode of Ψ and is deferred to a future revision. The qualitative features of the strong force — short range, confinement of vortex strand endpoints, and asymptotic freedom as quantum-pressure softening — follow from the same framework.

What the paper proposes as suggestive ansatz. For the weak force, the Rayleigh-jet cap formula $R_{\text{cap}} = \phi R^*$ recovers m_W to 1.8%. The cap equilibrium analysis (§5.4) shows this follows analytically from a chiral-shear bending stiffness $\lambda_{\text{bend}} = \phi^3/\alpha^3$; the local $\mathcal{L}_\perp$ curvature correction falls short by a factor 232, identifying the remaining open step as deriving λ_{bend} from the non-local (topological) structure of the cap. The formula is therefore no longer a bare fit to m_W, but a prediction conditional on the non-local bending stiffness.

What the paper offers as structural consistency. For gravity, the Bjerknes mechanism gives the inverse-square law, universal coupling, and time dilation as mechanical consequences of spherical wave dilution. The relation $G = \alpha_G \hbar c \alpha^2/(N_p^2 m_e^2)$ uses α_G from CODATA and matches the Paper I mass ladder to 0.6%. This is consistency between the gravity sector and the mass ladder, not a derivation of G; the 3D proton breather profile required to compute α_G from first principles remains the central open calculation. Gravitational and electromagnetic radiation are proposed to share the Goldstone (phase) channel of the medium and therefore propagate at the medium's sound speed c, consistent with GW170817; the relation between the chiral-shear sector at $c_\perp$ and the photon as observed is an open issue flagged in §3.9.

What the paper proposes as candidate geometry. The Higgs is recast as two distinct features: a structural amplitude mode of the LogSE Lagrangian (gapped at $m_H c^2$, present everywhere) and a derived denting event with energy $E_H = P_0 V^*$. Charged leptons are proposed as higher breathing harmonics of the toroidal electron vortex. Neutrinos are presented as a candidate torsional-radiation geometry; this candidate currently contradicts the established phenomenology in five ways (§5.7), and the section indicates where in the medium a neutrino-like object might live rather than offering a competing phenomenology.

What the paper raises as a programmatic question. The CP discussion observes that, in a medium that accumulates a record of every reconnection, no discrete symmetry of the dynamics need survive at the process level. This is a conceptual reframing. Whether it can be made quantitatively compatible with the observed precision of CP-even processes (small $\theta_{\rm QCD}$, small neutron EDM, predominantly CP-even behaviour across most kaon and B-meson observables) requires a suppression mechanism that is currently open.

What the paper identifies as the genuine open problems. The genuine hierarchy question in SSV — why the toroidal vortex ring has equilibrium radius $R^* \approx 137\,\xi$ while the knotted breather's core is $a_p \approx \xi/1836$, both solutions of the same field equations — is stated but not solved. Several apparently distinct open problems of the Standard Model — the W mass, the neutrino sector, the charged-lepton generations, and the CP observables — collapse, in this framework, into facets of a single 3D minimisation of the $\mathcal{L} + \mathcal{L}_\perp$ functional during a vortex reconnection event. The defect-spectrum hierarchy is an additional target of that minimisation. Whether the minimisation, when finally performed, produces the right numbers is the empirical test the framework sets for itself. That computation is the principal task left to subsequent work.

Scope. All claims above use the medium parameters first fixed by the mass spectrum of Paper I: the sound speed c, the fine-structure constant α, and the base mass scale $\mu_0 = m_e/\alpha$. No additional fields, symmetries, or coupling constants are introduced in this paper. The framework is presented as a continuum effective theory valid down to $\ell_{\rm grain} \equiv a_p \approx 2 \times 10^{-16}\,$m, the scale of the heaviest stable defect. What lies below this scale is a separate question, in the same sense that Navier–Stokes makes no claim about sub-molecular structure. The mean-field force-sector treatment of the present paper is the static envelope of a fuller picture in which the medium accumulates a record of every reconnection event. Paper III treats that record directly, identifying irreversible time and entropy with the growth of topological complexity.

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